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Relation between otitis media and sensorineural hearing loss: a systematic review

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Title:

Relation between otitis media and sensorineural hearing loss: a systematic review

Authors:
Hester B.E. Elzinga[1,*] & Hanneke D. van Oorschot[1,*] & Inge Stegeman [1,2, 3, 4] & Adriana L. Smit [1,2]

Corresponding author:

A.L. Smit, MD, PhD.

Department of Otorhinolaryngology and Head & Neck Surgery, P.O. Box 85500, 3508 GA Utrecht,

University Medical Center Utrecht, Utrecht, The Netherlands. E: a.l.smit-9@umcutrecht.nl / T: 0031-

88-7573291

Author affiliations:

1 Department of Otorhinolaryngology, Head and Neck Surgery, University Medical Center Utrecht, Utrecht, The Netherlands

2 Brain Center, University Medical Center Utrecht, Utrecht, The Netherlands

3 Department of Ophthalmology, University Medical Center Utrecht, Utrecht, The Netherlands

4 Epidemiology and Data Science, Amsterdam University Medical Centers, University of Amsterdam, Amsterdam, Netherlands.

*Both authors contributed equally to first authorship

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ABSTRACT

Objectives:

This systematic review summarizes the evidence on the correlation between CSOM/rAOM and SNHL.

Research methods:

PubMed, Embase and Cochrane Library databases were searched from inception to 15-January-2021. Two authors independently identified articles, extracted data and performed quality assessment for included studies. Studies comparing the sensorineural hearing levels of patients with a history of rAOM/CSOM for >3 months to a control-group were included.

Results:

Screening of 4168 articles lead to inclusion of two case-control studies (control-group: non-OM patients) and seven cohort-studies (control-group: contralateral ear). Quality assessment indicated considerable risk of bias in all studies. Reported populations varied (sample size 13-607, mean age 22-41.5 years, mean duration of disease 6.1-12.4 years). The OR for SNHL in the OM-group was 3.30-7.86 (95%CI 1.16-9.40, p<0.05) in cohort studies (n=2), and 0.05 (95%CI 0-0.78, p<0.05) in a case-control study. Mean/median bone conduction thresholds were respectively 1.19-32.21/0-10 dB higher on all frequencies (0.5-4kHz) for the OM-group in four cohort studies (p<0.05). Two other studies reported no statistical test outcomes.

Conclusion:

Due to the high risk of bias of included studies, effect estimates heterogeneity, and suboptimal research designs, no conclusion on the correlation between OM and SNHL can be made. It emphasizes the need for future prognostic studies.

Abstract word count: 246 words

Keywords: otitis media, OM, sensorineural hearing loss, SNHL, systematic review

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- A minimal otitis-exposure of three months was adhered to safeguard presence of inner-ear toxin-exposure before outcome assessment.
- A lack of prognostic studies complicates interpretation of effect estimates.
- Heterogeneity of definitions for otitis media and sensorineural hearing loss amongst studies complicate comparability.
- Recruitment of patients in an otolaryngology-setting with audiogram data introduces selection bias.
- Deficits in longitudinal design and methodology limit the quality of our results.

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INTRODUCTION

Otitis media (OM) has a great variety in presentation; it can occur transient as acute otitis media (AOM) and serous otitis media (SOM), or persistent as recurrent acute otitis media (rAOM) and chronic suppurative otitis media (CSOM).[1] It impacts lives worldwide, with global incidence rates estimated at 11% for AOM and 5% for CSOM, of which respectively 51% and 23% are children.[2] The exact incidences of the OM-subtypes vary between populations, depending on social, demographic and genetic factors for instance.[1]

During the acute inflammatory process, OM can lead to conductive hearing loss (HL).[3] In children this transient loss has not been related to a significant risk of delayed speech development or impaired school performance later on.[4,5] Though, the inflammatory process of the middle ear can be detrimental to the inner ear as shown in histological and animal studies.[6–8] During the inflammatory process, toxins can initiate damage to the round window membrane. This leads to diffusion of toxins to the inner ear and leakage of inner ear fluids, resulting in cochlear function loss; a permanent loss.[6–8] Till date, it remains unknown how many episodes of OM, and what duration of disease lead to inner ear hearing loss.

Given the age-related deterioration of inner-ear function, additional OM-related cochlear damage may become more prominent in later life. In observational studies, a statistical significant higher risk of developing permanent conductive and/or sensorineural hearing loss (SNHL) in adulthood has been demonstrated after experiencing CSOM at childhood age.[9–11] Several studies reported incidence numbers of SNHL up to 23-24% in CSOM patients.[12,13] In contrast to other risk factors for SNHL, OM is considered treatable by antibiotic or surgical interventions, which are available treatment options in most countries.[7] Though, up to this day, consensus on the long term hearing effects of OM has not been met, which complicates drawing conclusions and making recommendation for clinical care. [1,3]

Considering the above, we recognize a need for evidence on the effect of OM on sensorineural hearing levels in time. This knowledge could contribute to establish the best treatment guideline for this common condition, avoiding possible adverse long-term effects. We therefore conducted a systematic review to summarize the evidence on the correlation between rAOM or CSOM and SNHL in time.

METHODS

Methods - Patient and Public Involvement

No patients were involved.

Methods - reporting

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We applied the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline for this systematic review.[14]

Methods - Search strategy

On 15th of January 2021 a systematic search was performed in Embase, PubMed and Cochrane Library databases. No filters were applied. We used search terms and synonyms for the domain (CSOM, rAOM) and outcome (SNHL), including Emtree, and medical subject headings (MeSH) term fields. Used synonyms for the domain were otitis media, (middle) ear infection/inflammation, and abbreviations. Used synonyms for the outcome were hearing loss/impairment, deafness, cochlear dysfunction, hypoacusis, and inner ear impairment (Search syntax in Appendix 1). Although SOM did not apply to the study domain, it was included as a synonym to prevent exclusion of relevant articles during the initial search.

Methods - Study selection

Two authors blindly screened studies for inclusion (H.E., H.v.O.), and disagreements were resolved by discussion. Where disagreements could not be resolved, a third reviewer (D.S.) made the final decision. Prognostic, cohort, and cross-sectional studies were included. Cohort studies were categorized in retrospective or prospective study designs. Where this was methodologically unclear, the study was labeled as cohort study. Conference abstracts, or case reports were excluded. In case of an unavailable full text or abstract, the author was contacted to request full text. Reference screening and backwards citation tracking were performed to identify any missed literature. Rayyan was used as screening tool.[15] Full text screening and data extraction was independently performed by two authors (H.E., H.v.O.).

Original studies reporting on hearing outcome of patients of all ages with rAOM or CSOM and a control-group were included. Studies including participants with SNHL-related syndromes, cholesteatoma, or a history of ear surgery were excluded, as these factors independently cause SNHL. Articles that reported outcomes for several subgroups, of which only a selection met the inclusion criteria, data of that section was extracted and included in the analysis. CSOM was defined as a minimal presence of OM for three consecutive months or longer. Studies that included patients with OM of more than three months or had a minimal follow-up of three months were included. These criteria were set to safeguard a minimum period of three months in which inner-ear toxin exposure was present, before outcomes were assessed. Studies reporting effect estimates on hearing outcome for both the exposed and control-group were selected. To establish the relation between OM and SNHL, a shift of $> \underline{5}$ dB in hearing level between tone audiometric test results was found relevant considering the search range of 5 dB during these test procedures.[16]

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Methods – Data collection and analysis

Quality assessment of the studies

Critical appraisal of included studies was independently performed by two reviewers (H.E., H.v.O.) for validity and relevance. The Quality In Prognosis Studies (QUIPS) tool was used to assess risk of bias for prognostic study designs, and the Newcastle-Ottawa Scale (NOS) was used for cohort and case-control studies.[17,18] To score risk of bias consistently, criteria were predefined for both tools by the authors (Appendix 2). (H.E., H.v.O.) Using these tools, studies could be scored to contain low, moderate or high risk of bias (QUIPS), or to be of good, fair or poor quality (NOS). Differences in the appraisal of studies between the two reviewers in either tool were resolved through discussion.

Data and outcome extraction

Data on study and patient characteristics, follow-up, and BC hearing outcome of the included patients was independently extracted by two reviewers (H.E., H.v.O.). The primary outcome measure of cases and controls was reported in BC thresholds (dB) per frequency (0.5-4 kHz) and/or average hearing levels. Age, duration of disease, and the use of ototoxic ear drops were considered as confounding factors for outcomes of SNHL in groups.

Outcome measures and analyses

Odds ratios (OR) were either extracted directly or calculated by the authors (H.E., H.v.O.) if data was sufficient. If an OR could not be extracted or calculated, means and medians were extracted. A p-value of <0.05 was considered statistically significant. A meta-analysis was performed in case of methodologically homogenous studies.

RESULTS

Results - Search strategy and study collection

The search identified 4168 articles (Figure 1). After removal of duplicates, 3725 (89.4%) studies were screened based on title and abstract, as a result of which 3662 (98.3%) studies were excluded. After full-text analysis, 54 articles were excluded based on the outcome (n=22), study design (n=9), or by unavailability of the full text (n=23). No articles were received after correspondence with authors for full text. This resulted in a total of nine articles meeting all inclusion criteria.[19–27] Seven studies were cohort studies with a retrospective or prospective study design, or not further specified

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design.[19,20,22,24–26,28] Two studies were case-control studies.[23,27] No prognostic studies were found. Due to the methodological heterogeneity in included studies, no meta-analysis could be performed.

[INSERT FIGURE 1]

Results - Quality of the included articles

Critical appraisal of all studies was performed with the NOS (Table 1). The quality of included studies ranged from good (n=5),[19,20,26–28] to fair (n=2),[23,25] and poor (n=2).[22,24]

Selection

Selection bias was present in all studies, as patients were visiting an otolaryngology department before inclusion, none of the studies included a longitudinal cohort of the general population. Also, there was no objective demonstration that SNHL was not present before inclusion.[19–27] One study included self-reported diagnosis of OM instead of a doctor diagnosis.[23] Another study did not document information on recruitment of the cohort.[25]

Comparability

Most studies excluded patients with a history of ototoxic drug use.[19,22,24–27] Case-control studies matched cases and controls according to age and gender,[23,27] cohort studies used the contralateral ear as control-group.[19–22,24–26] Hearing outcomes were not corrected for known or unknow confounding factors in the included studies.[19–27]

Outcome

Rana et al. was the only study scoring low risk of bias for the outcome domain.[26] All other studies did not report information on missing data or loss to follow up.[19,20,23,25,27,28] Two studies could not guarantee a minimum duration of disease of three months for all their included patients.[22,24] Demographic information of included patients was retrieved from chart reviews. Otoscopy and audiometry were performed by trained personnel using conventional equipment.[19–27] Only the study of *de Azevedo* et al. reported that audiometry exams were carried out blinded in double by two independent professionals.[20]

[INSERT TABLE 1]

Results - Data extraction and study characteristics

Trial design and study sample

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Included studies were published between 1984 and 2019 and pertained to patients from Africa, Asia, Europe, North America and South America (Table 2). The sample sizes for the OM-group in cohort and case-control studies ranged from 70-607 (controls 70-534) and 13-137 (controls 13-137) respectively. Cohort studies used the contralateral ear as control-group.[19–22,24–26] Case-control studies used adults with <2 AOM episodes until 30 years of age[23] and volunteers of same age and gender[27] as control-group. Mean duration of disease varied between 6.1-12.4 years (range 0.2-42 years) in the four studies reporting it.[19,20,22,25] The mean age of patients was 22-41.5 years (range 0.25-73 years). Most study populations included both adults and children, except for two studies.[23,25] *Paparella* et al. did not describe the age of included patients, but excluded patients older than 60 years.[24]

Definitions for OM and SNHL

The included studies handled different inclusion criteria for the OM-group, namely: unilateral otorrhea >2 or >3 months with a tympanic membrane perforation (TMP)[19,22,26], chronic unilateral otorrhea[20], inflammation of the middle ear compartment (MEC) of >3 months[21], rAOM defined as ≥3 episodes of AOM during six months or ≥4 episodes during 12 months[23], a history of recurrent otorrhea within 3 years before data assembly[24], and chronic inflammation of the MEC with recurrent episodes of otorrhea through a TMP[27]. For *Krakau* et al., no criteria for inclusion in the OM-group were reported.[25]

Four studies defined their outcome as difference in BC threshold (dB) for cases and controls.[19,21,22,25] Four other studies defined a specific BC threshold level to establish cases with SNHL, namely: ≥30dB,[20] ≥15dB/≥30dB/any loss at 4 kHz,[24] ≥25dB,[26] and ≥20dB.[27] *Papp* et al. did not report a hearing outcome definition to be considered as SNHL.[23]

[INSERT TABLE 2]

Results - Outcomes

Three studies (2 cohort studies, 1 case-control study) reported odds ratios (OR) for the presence of SNHL after three months or longer exposure to OM.[20,24,27] (Table 3.1 and 3.2) The other six studies (5 cohort studies, 1 case-control studies) reported their outcome as a mean and/or median BC threshold for the OM-group and the control-group (Table 3.3 and 3.4).[19,21–23,25,26]

Outcomes in cohort studies

The OR for having SNHL could be extracted or calculated in two cohort studies comparing the contralateral to the diseased ear (Table 3.1).[20,24] Both reported higher ORs for the presence of SNHL after three months or longer exposure to OM. *Paparella* et al. reported an OR of 7.86 for a BC loss of

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≥15dB at ≥2 frequencies, 6.49 for BC loss of ≥30dB at ≥2 frequencies, and 4.41 for any BC loss at 4kHz (95% CI not reported).[24] For *de Azevedo* et al., the OR was 3.30, (95% CI 1.16-9.40, p=0.03).[20]

Five cohort studies reported a mean,[19,22,26] or both mean and median[21,25] for the BC hearing levels per frequency, stratified per study group (Table 3.3). Of those, four studies had a higher mean and/or median BC threshold in the OM-group compared to the control-group for all individually tested frequencies (0.5-4 kHz).[19,21,22,26] Mean and/or median differences between both groups per frequency ranged from 1.19-32.21 dB (p-value <0.05) for means and 0-10 dB (p-value <0.05) for medians.[19,21,22,26] *Papp* et al. reported the mean and median BC threshold for low (0.5-2kHz) and high (4 kHz) frequencies. The mean differences were higher for the OM-group compared to the control-group, though no statistical significance was determined.[25]

Outcomes in case-control studies

Subramaniam et al. reported data that was sufficient to calculate the OR (Table 3.2: 0.05 (95% CI 0.00-0.78, p=0.03) for SNHL when comparing the OM-patients to healthy volunteers, with SNHL defined as a BC loss of ≥20 dB at 0.25-4 Hz (Table 3.2).[27]

In one case-control study by *Krakau* et al. the authors reported median BC thresholds for the OM and control-group, stratified for the left and right ear (Table 3.4). The right ear showed a higher BC threshold (difference 10 dB, no p-value) for the OM-group compared to the control-group at 3 kHz. For all other frequencies, of both the right and left ear, the BC threshold difference between the groups was 5 dB or less.[23]

[INSERT TABLE 3.1, 3.2, 3.3, 3.4]

DISCUSSION

In this study, we described the results of a systematic search on the relation between rAOM or CSOM and the occurrence of SNHL in time. Two case-control studies (control-group: non-OM patients) and seven cohort-studies (control-group: contralateral ear) reported hearing outcomes for both the exposed and control-group.[19–26] The OR for the presence of SNHL after a minimal exposure to OM of three months was statistically significantly higher in two cohort studies,[20,24] but lower in one case-control study.[27] Studies in which no OR was provided or could be calculated, mean and/or median BC thresholds were statistically significantly higher in the OM-group in four out of six studies. Mean or median differences between both groups ranged between 1.19-32.21 dB and 0-10 dB respectively.[19,21,22,26] The other two studies reported differences of 0-5 dB in mean or median between OM-group and the contralateral ear[25] or controls, [23] though did not provide outcomes of statistical

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testing between groups. Outcomes of included studies should be interpreted cautiously by the small shift in hearing thresholds observed. Besides this, the heterogeneity between studies, the limited quality of included studies, and the lack of prognostic studies are all major limitations to draw conclusions about the effect of OM on SNHL.

None of the included articles assessed the effect of duration of the OM on SNHL. A longer period of exposure of the inner ear to inflammation, and therefore to toxins, could increase the portion of patients with SNHL.[29] Previous population studies analyzed the effect of the duration of the OM on hearing levels. *Thakur* et al. studied 100 cases (15-50 years of age) presenting with unilateral mucosal COM. They found an SNHL incidence of 23% (hearing level >20 dB), with a statistically significant positive correlation between duration of disease and SNHL.[12] A comparable SNHL incidence of 24% in 100 unilateral CSOM cases (aged 11-50 years) was published by *Kaur* et al, where the SNHL incidence (defined as hearing level >20 dB) increased with the duration of disease (up to >30 years).[13]

Besides exposure-time, one should take the following into consideration. Included studies were heterogenous due to varying sample sizes, age of study populations, definitions used for the OM-group, criteria for SNHL and method of reporting of outcome. Recruitment of patients was mostly done in an ENT-setting, and some studies handled absence of an audiogram as exclusion criterion which introduces selection or information bias. For example, *Kolo* et al. excluded patients if they had "incomplete" clinical records or no audiograms.[22] Age was accounted for by matching cases and controls or by using the contralateral ear. The use of the contralateral ear as non-affected control-group in cohort studies had the advantage of limiting the effect of inter-subject variations. On the other hand, if this ear suffered from infection prior to the study start, but was free of disease during the inclusion process, this could influence outcomes. However, in case-control studies, it could not be guaranteed that their control-group was free of previous OM-exposure.[22,24] For instance, *Krakau* et al. included a patient with 10 self-reported episodes of AOM in the control-group.[23] As stated above, no prognostic studies were included, and a meta-analysis could not be performed due to heterogeneity of reported results. A limitation of the present study is the considerable number of studies of which the full text could not be retrieved.

One could hypothesize that when inflammatory toxins are detrimental to the inner ear, the higher frequencies could be most affected considering their location at the base of the cochlea and in proximity of the inflammatory process in the middle ear. Four out of seven of our included studies reporting mean and/or median BC thresholds show a larger difference in hearing loss in the higher frequencies (up to 4 kHz). This is of interest for further studies as it could make people prone for hearing loss in the middle frequency by progression of a high frequency hearing loss by ageing.

The lack of longitudinal study designs and the selected sample of population studied so far hinders strong conclusions. In order to overcome this methodological flaw, studies with a longitudinal

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design, assessing a random sample of children and adults with and without OM for several decades is necessary.[30]

CONCLUSION

Considering the high risk of bias and the suboptimal research design of current evidence, no conclusions could be made about the relation between otitis media and SNHL in time. The high impact of SNHL on daily life, the high prevalence of OM in children and adults, and its potential effect on hearing, underpin the need for longitudinal studies assessing the effect of OM on hearing loss in time.

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AUTHOR CONTRIBUTIONS

H.E. and H.v.O. drafted the manuscript, searched and reviewed literature, assessed the quality of the studies, extracted data, and performed statistical analysis. All authors contributed to data review and interpretation and manuscript review. All authors have read and approved the final manuscript.

DATA STATEMENT

Data is accessible upon request.

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CONFLICT OF INTEREST

Authors declare there was no conflict of interest.

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TABLE AND FIGURE LEGENDS

Figure 1: Flow chart for included studies
Last date of search January 15th 2021 PRISMA 2009 flow diagram.[14]

Table 1: Risk of bias assessment of included studies according to the Newcastle-Ottawa Quality Assessment Scale
Risk of bias assessment of included studies according to the Newcastle-Ottawa Quality Assessment Scale[17]
F = Fair quality, G = Good quality, Max = maximum, P = Poor quality

Table 2: Characteristics of studies included in Systematic Review
The characteristics of the included studies are presented.
AOM = Acute Otitis Media, BC = Bone Conduction, BCT = Bone Conduction Threshold, CE = contralateral ear, COM = Chronic Otitis Media, CS = Cohort Study, dB = decibel, Hz = hertz, MEC = Middle Ear Compartment, OM = Otitis Media, PCS = Prospective Cohort Study, rAOM = recurrent Acute Otitis Media, RCS = Retrospective Cohort Study, SD = standard deviation, SNHL = Sensorineural Hearing Loss, TMP = Tympanic Membrane Perforation. NR = Not Reported
** Including 31% patients with cholesteatoma.*
*** Multicenter study reporting on subjects from Minnesota, Italy, Finland, Sweden, Korea and New York*[24]
****Research population +- 30 years of age with a history of rAOM during childhood*[23]
*****Averages in BC threshold of the contralateral (normal) ear were subtracted from that of the CSOM side.*[25]
****** The obtained thresholds were used to quantitatively assess the degree of hearing loss using Clark's (1981) modification of Goodman classification of severity of hearing loss (1965)*[27]

Table 3.1: Results of studies included in Systematic Review: Odds ratio of having SNHL when exposed to OM in cohort studies
Relevant results reported in odds ratio for having SNHL when exposed to OM compared to controls (CE)[20,24] in cohort studies
BC = Bone Conduction, CE = Contralateral Ear, CI = Confidence Interval, dB = decibel, kHz = kilo Hertz, OR = Odds Ratio
** Odds ratio was calculated manually using the reported data of the study by authors (H.E.; H.v.O.)*

Table 3.2: Results of studies included in Systematic Review: Odds ratio of having SNHL when exposed to OM in case-control studies
Relevant results reported in odds ratio for having SNHL when exposed to OM compared to controls (healthy "volunteers" of same age and gender)[27] in case-control studies
BC = Bone Conduction, CI = Confidence Interval, dB = decibel, kHz = kilo Hertz, OR = Odds Ratio
** Odds ratio was calculated manually using the reported data of the study by authors (H.E.; H.v.O.)*

Table 3.3: Results of studies included in Systematic Review: Mean and median BC thresholds of the OM-group and the contralateral ear
Relevant results of BC thresholds reported in mean and/or median of cohort studies included in Systematic Review are presented.
dB = decibel, Hz = Hertz OM = otitis media, Min = Minimum, Max = Maximum, SD = Standard Deviation

Table 3.4: Results of studies included in Systematic Review: Mean and median BC thresholds of the OM-group and the control-group
Relevant results of BC thresholds reported in mean and/or median of case-control studies included in Systematic Review are presented.
dB = decibel, Hz = Hertz OM = otitis media, Min = Minimum, Max = Maximum, SD = Standard Deviation

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Table 1: Risk of bias assessment of included studies according to the Newcastle-Ottawa Quality Assessment Scale

Author (year) / Domain:	Amali (2017)	De Azevedo (2007)	da Costa (2009)	Kolo (2011)	Krakau (2017)	Paparella (1984)	Papp (2003)	Rana (2019)	Subramaniam (2020)
Selection *Max: 4* ⋆	**3** ⋆	**3** ⋆	**3** ⋆	**3** ⋆	**2** ⋆	**3** ⋆	**2** ⋆	**3** ⋆	**3** ⋆
Comparability *Max: 2* ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆
Outcome *Max: 3* ⋆	**2** ⋆	**2** ⋆	**2** ⋆	**1** ⋆	**2** ⋆	**1** ⋆	**2** ⋆	**3** ⋆	**2** ⋆
Quality assessment	**G**	**G**	**G**	**P**	**F**	**P**	**F**	**G**	**G**

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Table 2: Characteristics of studies included in Systematic Review

Author (country, year of publication)	Study design	Sample size OM group (control group)	Control group	Mean duration of disease in years ±SD (min-max)	Mean age in years ±SD (min-max)	OM-group definition	SNHL outcome definition
Amali (Iran, 2017)	RCS	70 (70)	CE	6.5±7.1 (0.25-30)	37.9±15 (0.25-30)	Unilateral otorrhea >3 months and TMP	Difference in BC threshold for cases and controls
de Azevedo (Brazil, 2007)	RCS	115 (115)	CE	12.4±10.9 (1-42)	26.3±15.7 (4-59)	Chronic unilateral otorrhea	BC loss of ≥30dB
da Costa (Brazil, 2009)	RCS	150*(150)	CE	NR	22±13 (3-55)	Inflammation of MEC >3 months	Difference in BC threshold for cases and controls
Kolo (Nigeria, 2011)	RCS	69 (69)	CE	6.1±6.4 (0.2-20)	28.9±18.6 (4-73)	Unilateral otorrhea >2 months and TMP	Difference in BC threshold for cases and controls
Krakau (Sweden, 2017)	Case-Control	13 (15)	<2 AOM episodes until 30 years of age	***	30.1–31.8 ***	rAOM: ≥3 episodes of AOM during six months or ≥4 episodes during 12 months	NR
*Paparella (Multicenter** *, 1984)*	CS	607 (534)	CE	NR	NR	History of recurrent otorrhea <3 years of selection date	BC loss of ≥15dB, ≥30dB, or any loss at 4kHz
Papp (Hungary, 2003)	RCS	121 (121)	CE ****	10.0±9.6	36.9±14.3	NR	Difference in BC threshold for cases and controls
Rana (India, 2019)	PCS	146 (146)	CE	NR	41.5±8.9 (10-60)	(Unilateral) otorrhea >3 months and TMP	BC threshold of ≥25 dB
Subramaniam (Iran, 2020)	Case-Control	137 (137)	Volunteers of same age and gender	NR	(11-45)	Chronic inflammation of MEC with recurrent episodes of otorrhea through a TMP	BC loss of ≥20 dB at 0.25-4 kHz*****

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Table 3.1: Results of studies included in Systematic Review: Odds ratio of having SNHL when exposed to OM in cohort studies

Author (year)	Control group	SNHL outcome description	OR (95% CI)	P-value
De Azevedo* (2007)	CE	BC loss of ≥30dB	3.30 (1.16-9.40)	0.03
Paparella (1984)	CE	BC loss ≥15dB at ≥2 frequencies BC loss ≥30dB at ≥2 frequencies BC loss a 4kHz only	7.86 (NR) 6.49 (NR) 4.41 (NR)	<0.05 <0.05 <0.05

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Table 3.2: Results of studies included in Systematic Review: Odds ratio of having SNHL when exposed to OM in case-control studies

Author (year)	Control group	SNHL outcome description	OR (95% CI)	P-value
Subramaniam* (2020)	Volunteers of same age and gender	BC loss of ≥20 dB at 0.25-4 kHz	0.05 (0.00-0.78)	0.03

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Table 3.3: Results of studies included in Systematic Review: Mean and median BC thresholds of the OM-group and the contralateral ear

Author (year)	Effect estimate reported in dB: Mean ± SD Median (min/max)		Frequency						
			0.25 kHz	0.5 kHz	1 kHz	1.5 kHz	2 kHz	3 kHz	4 kHz
Amali (2017)	Mean	OM-Group	-	17.57 ± 13.93	18.36 ± 16.30	-	21.71 ± 16.92	-	21.71 ± 16.92
		Control-group	-	10.57 ± 9.07	10.81 ± 9.96	-	13.21 ± 12.10	-	18.29 ± 14.77
		P-value	-	<0.001	<0.001	-	<0.001	-	<0.001
da Costa (2009)	Mean	OM-Group	-	6.63 ± 9.06	6.88 ± 9.01	-	8.81 ± 8.28	12.44 ± 9.41	12.56 ± 10.40
		Control-group	-	5.44 ± 5.69	4.69 ± 4.80	-	5.25 ± 4.96	6.00 ± 4.73	5.69 ± 5.20
		P-value	-	0.001	0.001	-	<0.0001	<0.0001	<0.0001
	Median	OM-Group		5 (0/50)	10 (0/50)	-	10 (0/40)	15 (0/40)	15 (0/50)
		Control-group		5 (0/20)	5 (0/15)	-	5 (0/20)	5 (0/20)	5 (0/20)
Kolo (2011)	Mean	OM-Group	-	32.46 ± 14.587	36.06 ± 14.140	-	42.18 ± 15.183	-	45.511 ± 15.384
		Control-group	-	10.14 ± 4.470	10.14 ±5.208	-	9.97 ± 4.585	-	10.99 ± 3.930
		P-value	-	<0.05	<0.05	-	<0.05	-	<0.05
Papp (2003)	Mean	OM-Group	-	10.4 ± 4.89				-	11.9 ± 6.78
		Control-group	-	4.49 ± 3.947				-	3.10 ± 4.510
		P-value	-	NR				-	NR
	Median	OM-Group		10 (3.3/30)				-	3.34 (0/21.67)
		Control-group		10 (2.5/35)				-	2.5 (0/30)
Rana (2019)	Mean	OM-Group	-	23.50 ± 4.5	24.06 ± 2.4	-	36.18 ± 5.6	-	40.50 ± 6.4
		Control-group	-	18.40 (SD 4.2)	16.6 ± 2.1	-	9.90 ± 4.1	-	11.00 ± 5.2
		P-value	-	<0.001	<0.001	-	<0.001	-	<0.001

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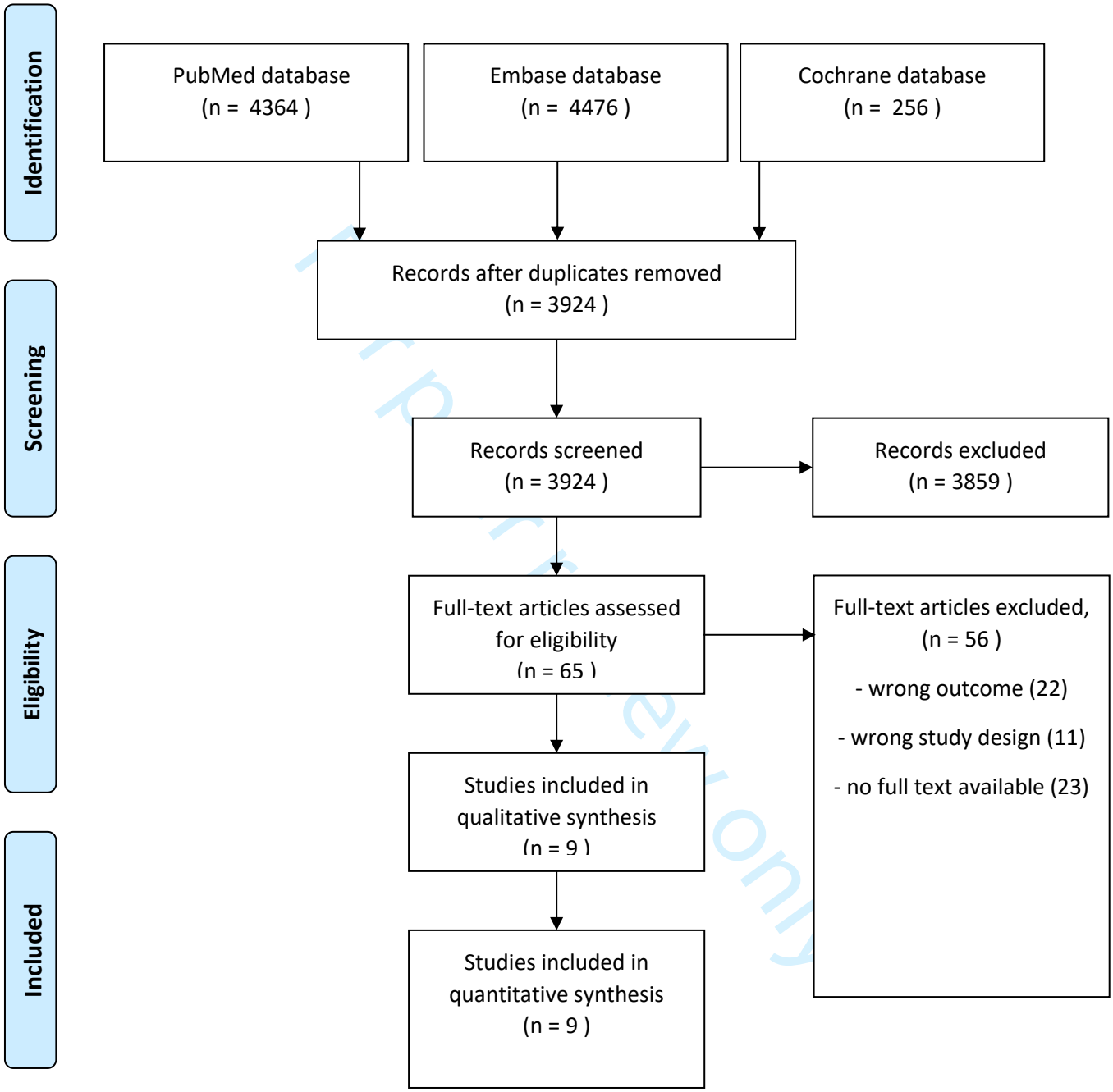
Table 3.4: Results of studies included in Systematic Review: Mean and median BC thresholds of the OM-group and the control-group

Author (year)	Effect estimate reported in dB: Mean ± SD Median (min/max)		Frequency						
			0.25 kHz	0.5 kHz	1 kHz	1.5 kHz	2 kHz	3 kHz	4 kHz
Krakau (2017)	Median	OM-Group Right ear	5 (0 /20)	5 (0/15)	0 (-5/10)	0 (-10/25)	10 (0/20)	10 (-5/25)	0 (-5/25)
		Control-group Right ear	0 (-5/10)	5 (-5/10)	0 (-10/10)	0 (-10/15)	10 (-10/40)	0 (-5/10)	5 (-10/20)
		OM-Group Left ear	5 (0/20)	5 (0/20)	0 (-10/10)	0 (-10/10)	10 (-5/45)	5 (-5/25)	5 (-5/25)
		Control-group Left ear	5 (-5/10)	5 (0/15)	0 (-5/10)	5 (-5/20)	10 (0/45)	0 (-5/10)	0 (-10/15)
		P-value	NR						

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Figure 1: Flow chart for included studies



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APPENDIX 1: Search syntax

PUBMED SEARCH
"Otitis Media"[Mesh] OR otitis media[Title/Abstract] OR ear infection*[Title/Abstract] OR ear inflammation*[Title/Abstract] OR otiti* media[Title/Abstract] OR middle ear inf*[Title/Abstract] OR OME[Title/Abstract] OR OMA[Title/Abstract] OR CSOM[Title/Abstract] OR otitides[Title/Abstract] OR ACSOM[Title/Abstract] AND "Hearing Loss"[Mesh] OR hearing loss*[Title/Abstract] OR deafness[Title/Abstract] OR cochlear dysfunction*[Title/Abstract] OR hypoacusis[Title/Abstract] OR hearing impairment[Title/Abstract] OR inner ear impairment[Title/Abstract]
EMBASE SEARCH
'otitis media'/exp OR 'otitis media':ti,ab,kw OR 'ear infection*':ti,ab,kw OR 'ear inflammation*':ti,ab,kw OR 'otiti* media':ti,ab,kw OR 'middle ear inf*':ti,ab,kw OR 'OME':ti,ab,kw OR 'OMA':ti,ab,kw OR 'CSOM':ti,ab,kw OR 'otitides':ti,ab,kw OR 'ACSOM':ti,ab,kw AND 'inner ear hearing loss'/exp OR 'hearing loss*':ti,ab,kw OR 'deafness':ti,ab,kw OR 'cochlear dysfunction*':ti,ab,kw OR 'hypoacusis':ti,ab,kw OR 'hearing impairment':ti,ab,kw OR 'inner ear impairment':ti,ab,kw
COCHRANE SEARCH
"otitis media":ti,ab,kw OR "ear infection*":ti,ab,kw OR "ear inflammation*":ti,ab,kw OR "otiti* media":ti,ab,kw OR "middle ear inf*":ti,ab,kw OR "OME":ti,ab,kw OR "OMA":ti,ab,kw OR "CSOM":ti,ab,kw OR "otitides":ti,ab,kw OR "ACSOM":ti,ab,kw AND "hearing loss*":ti,ab,kw OR "deafness":ti,ab,kw OR "cochlear dysfunction*":ti,ab,kw OR "hypoacusis":ti,ab,kw OR "hearing impairment":ti,ab,kw OR "inner ear impairment":ti,ab,kw

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APPENDIX 2: predefined risk of bias consistently criteria

The Quality In Prognosis Studies tool (QUIPS)

Domain	Criteria
Study Participation	- In-/exclusion criteria: if only including patients with audiograms, choose no (selection bias). - Baseline characteristics: when most important factors are described but no table is given, choose yes.
Study attrition	No additional criteria
Prognostic factor measurement	- Definition of the prognostic factor: must contain >3 months.
Outcome measurement	No additional criteria
Study confounding	- Confounders: age, duration of disease, (prior) use of ototoxic ear drugs.
Statistical analysis and reporting	No additional criteria

Newcastle-Ottawa Score (NOS)

Category	Criteria
• Selection	No additional criteria
• Comparability	- Total of 1 star when accounting for age (cohort: contralateral ear; case-control: matching). - Total of 2 stars when also accounting for duration of disease and prior use of ear drugs.
• Outcome	- Choose record linkage when outcome information was derived from chart reviews. - Follow up was long enough when duration of disease was >3 months.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	5
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7-8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	8-9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	8-9
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	8-9
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	10

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Keywords:	OTOLARYNGOLOGY, Audiology < OTOLARYNGOLOGY, Paediatric otolaryngology < OTOLARYNGOLOGY

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Title:

Relation between otitis media and sensorineural hearing loss: a systematic review

Authors:
Hester B.E. Elzinga[1,*] & Hanneke D. van Oorschot[1,*] & Inge Stegeman [1,2, 3, 4] & Adriana L. Smit [1,2]

Corresponding author:

A.L. Smit, MD, PhD.

Department of Otorhinolaryngology and Head & Neck Surgery, P.O. Box 85500, 3508 GA Utrecht,

University Medical Center Utrecht, Utrecht, The Netherlands. E: a.l.smit-9@umcutrecht.nl / T: 0031-

88-7573291

Author affiliations:

1 Department of Otorhinolaryngology, Head and Neck Surgery, University Medical Center Utrecht, Utrecht, The Netherlands

2 Brain Center, University Medical Center Utrecht, Utrecht, The Netherlands

3 Department of Ophthalmology, University Medical Center Utrecht, Utrecht, The Netherlands

4 Epidemiology and Data Science, Amsterdam University Medical Centers, University of Amsterdam, Amsterdam, Netherlands.

*Both authors contributed equally to first authorship

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ABSTRACT

Objectives:

This systematic review summarizes the evidence on the correlation between CSOM/rAOM and SNHL.

Research methods:

PubMed, Embase and Cochrane Library databases were searched from inception to 15-January-2021. Two authors independently identified articles, extracted data and performed quality assessment for included studies. Studies comparing the sensorineural hearing levels of patients with a history of rAOM/CSOM for >3 months to a control-group were included.

Results:

Screening of 4168 articles lead to inclusion of two case-control studies (control-group: non-OM patients) and seven cohort-studies (control-group: contralateral ear). Quality assessment indicated considerable risk of bias in all studies. Reported populations varied (sample size 13-607, mean age 22-41.5 years, mean duration of disease 6.1-12.4 years). The OR for SNHL in the OM-group was 3.30-7.86 (95%CI 1.16-9.40, p<0.05) in cohort studies (n=2), and 0.05 (95%CI 0-0.78, p<0.05) in a case-control study. Mean/median bone conduction thresholds were respectively 1.19-32.21/0-10 dB higher on all frequencies (0.5-4kHz) for the OM-group in four cohort studies (p<0.05). Two other studies reported no statistical test outcomes.

Conclusion:

Due to the high risk of bias of included studies, effect estimates heterogeneity, and suboptimal research designs, no conclusion on the correlation between OM and SNHL can be made. It emphasizes the need for future prognostic studies.

Abstract word count: 246 words

Keywords: otitis media, OM, sensorineural hearing loss, SNHL, systematic review

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- A minimal otitis-exposure of three months was adhered to safeguard presence of inner-ear toxin-exposure before outcome assessment.
- A lack of prognostic studies complicates interpretation of effect estimates.
- Heterogeneity of definitions for otitis media and sensorineural hearing loss amongst studies complicate comparability.
- Recruitment of patients in an otolaryngology-setting with audiogram data introduces selection bias.
- Deficits in longitudinal design and methodology limit the quality of our results.

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INTRODUCTION

Otitis media (OM) has a great variety in presentation; it can occur transient as acute otitis media (AOM) and serous otitis media (SOM), or persistent as recurrent acute otitis media (rAOM) and chronic suppurative otitis media (CSOM).[1] It impacts lives worldwide, with global incidence rates estimated at 11% for AOM and 5% for CSOM, of which respectively 51% and 23% are children.[2] The exact incidences of the OM-subtypes vary between populations, depending on social, demographic and genetic factors for instance.[1]

During the acute inflammatory process, OM can lead to conductive hearing loss (HL).[3] In children this transient loss during the acute phase has not been related to a significant risk of delayed speech development or impaired school performance later on.[4,5] Though, the inflammatory process of the middle ear can be detrimental to the inner ear as shown in histological and animal studies.[6–8] During the inflammatory process, toxins can initiate damage to the round window membrane. This leads to diffusion of toxins to the inner ear and leakage of inner ear fluids, resulting in cochlear function loss; a permanent loss.[6–8] Till date, it remains unknown how many episodes of OM, and what duration of disease lead to inner ear hearing loss.

Given the age-related deterioration of inner-ear function, additional OM-related cochlear damage may become more prominent in later life. In observational studies, a statistical significant higher risk of developing permanent conductive and/or sensorineural hearing loss (SNHL) in adulthood has been demonstrated after experiencing CSOM at childhood age.[9–11] Several studies reported incidence numbers of SNHL up to 23-24% in CSOM patients.[12,13] In contrast to other risk factors for SNHL, OM is considered treatable by antibiotic or surgical interventions, which are available treatment options in most countries.[7] Though, up to this day, consensus on the long term hearing effects of OM has not been met, which complicates drawing conclusions and making recommendation for clinical care. [1,3]

Considering the above, we recognize a need for evidence on the effect of OM on sensorineural hearing levels in time. This knowledge could contribute to establish the best treatment guideline for this common condition, avoiding possible adverse long-term effects. We therefore conducted a systematic review to summarize the evidence on the correlation between rAOM or CSOM and SNHL in time.

METHODS

Methods - Patient and Public Involvement

No patients were involved.

Methods - reporting

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We applied the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline for this systematic review.[14]

Methods - Search strategy

On 15th of January 2021 a systematic search was performed in Embase, PubMed and Cochrane Library databases. No filters were applied. We used search terms and synonyms for the domain (CSOM, rAOM) and outcome (SNHL), including Emtree, and medical subject headings (MeSH) term fields. Used synonyms for the domain were otitis media, (middle) ear infection/inflammation, and abbreviations. Used synonyms for the outcome were hearing loss/impairment, deafness, cochlear dysfunction, hypoacusis, and inner ear impairment (Search syntax in Appendix 1). Although SOM did not apply to the study domain, it was included as a synonym to prevent exclusion of relevant articles during the initial search.

Methods - Study selection

Two authors blindly screened studies for inclusion (H.E., H.v.O.), and disagreements were resolved by discussion. Where disagreements could not be resolved, a third reviewer (D.S.) made the final decision. Prognostic, cohort, and cross-sectional studies were included. Cohort studies were categorized in retrospective or prospective study designs. Where this was methodologically unclear, the study was labeled as cohort study. Conference abstracts, or case reports were excluded. In case of an unavailable full text or abstract, the author was contacted to request full text. Reference screening and backwards citation tracking were performed to identify any missed literature. Rayyan was used as screening tool.[15] Full text screening and data extraction was independently performed by two authors (H.E., H.v.O.).

Original studies reporting on hearing outcome of patients of all ages with rAOM or CSOM and a control-group were included. Studies including participants with SNHL-related syndromes, cholesteatoma, or a history of ear surgery were excluded, as these factors independently cause SNHL. Articles that reported outcomes for several subgroups, of which only a selection met the inclusion criteria, data of that section was extracted and included in the analysis. CSOM was defined as a minimal presence of OM for three consecutive months or longer. Studies that included patients with OM of more than three months or had a minimal follow-up of three months were included. These criteria were set to safeguard a minimum period of three months in which inner-ear toxin exposure was present, before outcomes were assessed. Studies reporting effect estimates on hearing outcome for both the exposed and control-group were selected. To establish the relation between OM and SNHL, a shift of $> \underline{5}$ dB in hearing level between tone audiometric test results was found relevant considering the search range of 5 dB during these test procedures.[16]

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Methods – Data collection and analysis

Quality assessment of the studies

Critical appraisal of included studies was independently performed by two reviewers (H.E., H.v.O.) for validity and relevance. The Quality In Prognosis Studies (QUIPS) tool was used to assess risk of bias for prognostic study designs, and the Newcastle-Ottawa Scale (NOS) was used for cohort and case-control studies.[17,18] To score risk of bias consistently, criteria were predefined for both tools by the authors (Appendix 2). (H.E., H.v.O.) Using these tools, studies could be scored to contain low, moderate or high risk of bias (QUIPS), or to be of good, fair or poor quality (NOS). Differences in the appraisal of studies between the two reviewers in either tool were resolved through discussion.

Data and outcome extraction

Data on study and patient characteristics, follow-up, and BC hearing outcome of the included patients was independently extracted by two reviewers (H.E., H.v.O.). The primary outcome measure of cases and controls was reported in BC thresholds (dB) per frequency (0.5-4 kHz) and/or average hearing levels. Age, duration of disease, and the use of ototoxic ear drops were considered as confounding factors for outcomes of SNHL in groups.

Outcome measures and analyses

Odds ratios (OR) were either extracted directly or calculated by the authors (H.E., H.v.O.) if data was sufficient. If an OR could not be extracted or calculated, means and medians were extracted. A p-value of <0.05 was considered statistically significant. A meta-analysis was performed in case of methodologically homogenous studies.

RESULTS

Results - Search strategy and study collection

The search identified 4168 articles (Figure 1). After removal of duplicates, 3725 (89.4%) studies were screened based on title and abstract, as a result of which 3662 (98.3%) studies were excluded. After full-text analysis, 54 articles were excluded based on the outcome (n=22), study design (n=9), or by unavailability of the full text (n=23). No articles were received after correspondence with authors for full text. This resulted in a total of nine articles meeting all inclusion criteria.[19–27] Seven studies were cohort studies with a retrospective or prospective study design, or not further specified

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design.[19,20,22,24–26,28] Two studies were case-control studies.[23,27] No prognostic studies were found. Due to the methodological heterogeneity in included studies, no meta-analysis could be performed.

[INSERT FIGURE 1]

Results - Quality of the included articles

Critical appraisal of all studies was performed with the NOS (Table 1). The quality of included studies ranged from good (n=5),[19,20,26–28] to fair (n=2),[23,25] and poor (n=2).[22,24]

Selection

Selection bias was present in all studies, as patients were visiting an otolaryngology department before inclusion, none of the studies included a longitudinal cohort of the general population. Also, there was no objective demonstration that SNHL was not present before inclusion.[19–27] One study included self-reported diagnosis of OM instead of a doctor diagnosis.[23] Another study did not document information on recruitment of the cohort.[25]

Comparability

Most studies excluded patients with a history of ototoxic drug use.[19,22,24–27] Case-control studies matched cases and controls according to age and gender,[23,27] cohort studies used the contralateral ear as control-group.[19–22,24–26] Hearing outcomes were not corrected for known or unknow confounding factors in the included studies.[19–27]

Outcome

Rana et al. was the only study scoring low risk of bias for the outcome domain.[26] All other studies did not report information on missing data or loss to follow up.[19,20,23,25,27,28] Two studies could not guarantee a minimum duration of disease of three months for all their included patients.[22,24] Demographic information of included patients was retrieved from chart reviews. Otoscopy and audiometry were performed by trained personnel using conventional equipment.[19–27] Only the study of *de Azevedo* et al. reported that audiometry exams were carried out blinded in double by two independent professionals.[20]

[INSERT TABLE 1]

Results - Data extraction and study characteristics

Trial design and study sample

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Included studies were published between 1984 and 2019 and pertained to patients from Africa, Asia, Europe, North America and South America (Table 2). The sample sizes for the OM-group in cohort and case-control studies ranged from 70-607 (controls 70-534) and 13-137 (controls 13-137) respectively. Cohort studies used the contralateral ear as control-group.[19–22,24–26] Case-control studies used adults with <2 AOM episodes until 30 years of age[23] and volunteers of same age and gender[27] as control-group. Mean duration of disease varied between 6.1-12.4 years (range 0.2-42 years) in the four studies reporting it.[19,20,22,25] The mean age of patients was 22-41.5 years (range 0.25-73 years). Most study populations included both adults and children, except for two studies.[23,25] *Paparella* et al. did not describe the age of included patients, but excluded patients older than 60 years.[24]

Definitions for OM and SNHL

The included studies handled different inclusion criteria for the OM-group, namely: unilateral otorrhea >2 or >3 months with a tympanic membrane perforation (TMP)[19,22,26], chronic unilateral otorrhea[20], inflammation of the middle ear compartment (MEC) of >3 months[21], rAOM defined as ≥3 episodes of AOM during six months or ≥4 episodes during 12 months[23], a history of recurrent otorrhea within 3 years before data assembly[24], and chronic inflammation of the MEC with recurrent episodes of otorrhea through a TMP[27]. For *Krakau* et al., no criteria for inclusion in the OM-group were reported.[25]

Four studies defined their outcome as difference in BC threshold (dB) for cases and controls.[19,21,22,25] Four other studies defined a specific BC threshold level to establish cases with SNHL, namely: ≥30dB,[20] ≥15dB/≥30dB/any loss at 4 kHz,[24] ≥25dB,[26] and ≥20dB.[27] *Papp* et al. did not report a hearing outcome definition to be considered as SNHL.[23]

[INSERT TABLE 2]

Results - Outcomes

Three studies (2 cohort studies, 1 case-control study) reported odds ratios (OR) for the presence of SNHL after three months or longer exposure to OM.[20,24,27] (Table 3.1 and 3.2) The other six studies (5 cohort studies, 1 case-control studies) reported their outcome as a mean and/or median BC threshold for the OM-group and the control-group (Table 3.3 and 3.4).[19,21–23,25,26]

Outcomes in cohort studies

The OR for having SNHL could be extracted or calculated in two cohort studies comparing the contralateral to the diseased ear (Table 3.1).[20,24] Both reported higher ORs for the presence of SNHL after three months or longer exposure to OM. *Paparella* et al. reported an OR of 7.86 for a BC loss of

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≥15dB at ≥2 frequencies, 6.49 for BC loss of ≥30dB at ≥2 frequencies, and 4.41 for any BC loss at 4kHz (95% CI not reported).[24] For *de Azevedo* et al., the OR was 3.30, (95% CI 1.16-9.40, p=0.03).[20]

Five cohort studies reported a mean,[19,22,26] or both mean and median[21,25] for the BC hearing levels per frequency, stratified per study group (Table 3.3). Of those, four studies had a higher mean and/or median BC threshold in the OM-group compared to the control-group for all individually tested frequencies (0.5-4 kHz).[19,21,22,26] Mean and/or median differences between both groups per frequency ranged from 1.19-32.21 dB (p-value <0.05) for means and 0-10 dB (p-value <0.05) for medians.[19,21,22,26] *Papp* et al. reported the mean and median BC threshold for low (0.5-2kHz) and high (4 kHz) frequencies. The mean differences were higher for the OM-group compared to the control-group, though no statistical significance was determined.[25]

Outcomes in case-control studies

Subramaniam et al. reported data that was sufficient to calculate the OR (Table 3.2: 0.05 (95% CI 0.00-0.78, p=0.03) for SNHL when comparing the OM-patients to healthy volunteers, with SNHL defined as a BC loss of ≥20 dB at 0.25-4 Hz (Table 3.2).[27]

In one case-control study by *Krakau* et al. the authors reported median BC thresholds for the OM and control-group, stratified for the left and right ear (Table 3.4). The right ear showed a higher BC threshold (difference 10 dB, no p-value) for the OM-group compared to the control-group at 3 kHz. For all other frequencies, of both the right and left ear, the BC threshold difference between the groups was 5 dB or less.[23]

[INSERT TABLE 3.1, 3.2, 3.3, 3.4]

DISCUSSION

In this study, we described the results of a systematic search on the relation between rAOM or CSOM and the occurrence of SNHL in time. Two case-control studies (control-group: non-OM patients) and seven cohort-studies (control-group: contralateral ear) reported hearing outcomes for both the exposed and control-group.[19–26] The OR for the presence of SNHL after a minimal exposure to OM of three months was statistically significantly higher in two cohort studies,[20,24] but lower in one case-control study.[27] Studies in which no OR was provided or could be calculated, mean and/or median BC thresholds were statistically significantly higher in the OM-group in four out of six studies. Mean or median differences between both groups ranged between 1.19-32.21 dB and 0-10 dB respectively.[19,21,22,26] The other two studies reported differences of 0-5 dB in mean or median between OM-group and the contralateral ear[25] or controls, [23] though did not provide outcomes of statistical

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testing between groups. Outcomes of included studies should be interpreted cautiously by the small shift in hearing thresholds observed. Besides this, the heterogeneity between studies, the limited quality of included studies, and the lack of prognostic studies are all major limitations to draw conclusions about the effect of OM on SNHL.

None of the included articles assessed the effect of duration of the OM on SNHL. A longer period of exposure of the inner ear to inflammation, and therefore to toxins, could increase the portion of patients with SNHL.[29] Previous population studies analyzed the effect of the duration of the OM on hearing levels. *Thakur* et al. studied 100 cases (15-50 years of age) presenting with unilateral mucosal COM. They found an SNHL incidence of 23% (hearing level >20 dB), with a statistically significant positive correlation between duration of disease and SNHL.[12] A comparable SNHL incidence of 24% in 100 unilateral CSOM cases (aged 11-50 years) was published by *Kaur* et al, where the SNHL incidence (defined as hearing level >20 dB) increased with the duration of disease (up to >30 years).[13]

Besides exposure-time, one should take the following into consideration. Included studies were heterogenous due to varying sample sizes, age of study populations, definitions used for the OM-group, criteria for SNHL and method of reporting of outcome. Recruitment of patients was mostly done in an ENT-setting, and some studies handled absence of an audiogram as exclusion criterion which introduces selection or information bias. For example, *Kolo* et al. excluded patients if they had "incomplete" clinical records or no audiograms.[22] Age was accounted for by matching cases and controls or by using the contralateral ear. The use of the contralateral ear as non-affected control-group in cohort studies had the advantage of limiting the effect of inter-subject variations. On the other hand, if this ear suffered from infection prior to the study start, but was free of disease during the inclusion process, this could influence outcomes. However, in case-control studies, it could not be guaranteed that their control-group was free of previous OM-exposure.[22,24] For instance, *Krakau* et al. included a patient with 10 self-reported episodes of AOM in the control-group.[23] As stated above, no prognostic studies were included, and a meta-analysis could not be performed due to heterogeneity of reported results. A limitation of the present study is the considerable number of studies of which the full text could not be retrieved.

One could hypothesize that when inflammatory toxins are detrimental to the inner ear, the higher frequencies could be most affected considering their location at the base of the cochlea and in proximity of the inflammatory process in the middle ear. Four out of seven of our included studies reporting mean and/or median BC thresholds show a larger difference in hearing loss in the higher frequencies (up to 4 kHz). This is of interest for further studies as it could make people prone for hearing loss in the middle frequency by progression of a high frequency hearing loss by ageing.

The lack of longitudinal study designs and the selected sample of population studied so far hinders strong conclusions. In order to overcome this methodological flaw, studies with a longitudinal

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design, assessing a random sample of children and adults with and without OM for several decades is necessary.[30]

CONCLUSION

Considering the high risk of bias and the suboptimal research design of current evidence, no conclusions could be made about the relation between otitis media and SNHL in time. The high impact of SNHL on daily life, the high prevalence of OM in children and adults, and its potential effect on hearing, underpin the need for longitudinal studies assessing the effect of OM on hearing loss in time.

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AUTHOR CONTRIBUTIONS

I.S. and A.S. were involved in conception of the study. H.E. and H.v.O. drafted the manuscript, searched and reviewed literature, assessed the quality of the studies, extracted data, and performed statistical analysis. H.E., H.v.O., I.S. and A.S. contributed to data review and interpretation and manuscript review. H.E., H.v.O., I.S. and A.S. have read and approved the final manuscript.

DATA STATEMENT

Data is accessible upon request.

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CONFLICT OF INTEREST

Authors declare there was no conflict of interest.

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TABLE AND FIGURE LEGENDS

Figure 1: Flow chart for included studies
Last date of search January 15th 2021 PRISMA 2009 flow diagram.[14]

Table 1: Risk of bias assessment of included studies according to the Newcastle-Ottawa Quality Assessment Scale
Risk of bias assessment of included studies according to the Newcastle-Ottawa Quality Assessment Scale[17]
F = Fair quality, G = Good quality, Max = maximum, P = Poor quality

Table 2: Characteristics of studies included in Systematic Review
The characteristics of the included studies are presented.
AOM = Acute Otitis Media, BC = Bone Conduction, BCT = Bone Conduction Threshold, CE = contralateral ear, COM = Chronic Otitis Media, CS = Cohort Study, dB = decibel, Hz = hertz, MEC = Middle Ear Compartment, OM = Otitis Media, PCS = Prospective Cohort Study, rAOM = recurrent Acute Otitis Media, RCS = Retrospective Cohort Study, SD = standard deviation, SNHL = Sensorineural Hearing Loss, TMP = Tympanic Membrane Perforation. NR = Not Reported
** Including 31% patients with cholesteatoma.*
*** Multicenter study reporting on subjects from Minnesota, Italy, Finland, Sweden, Korea and New York*[24]
****Research population +- 30 years of age with a history of rAOM during childhood*[23]
*****Averages in BC threshold of the contralateral (normal) ear were subtracted from that of the CSOM side.*[25]
****** The obtained thresholds were used to quantitatively assess the degree of hearing loss using Clark's (1981) modification of Goodman classification of severity of hearing loss (1965)*[27]

Table 3.1: Results of studies included in Systematic Review: Odds ratio of having SNHL when exposed to OM in cohort studies
Relevant results reported in odds ratio for having SNHL when exposed to OM compared to controls (CE)[20,24] in cohort studies
BC = Bone Conduction, CE = Contralateral Ear, CI = Confidence Interval, dB = decibel, kHz = kilo Hertz, OR = Odds Ratio
** Odds ratio was calculated manually using the reported data of the study by authors (H.E.; H.v.O.)*

Table 3.2: Results of studies included in Systematic Review: Odds ratio of having SNHL when exposed to OM in case-control studies
Relevant results reported in odds ratio for having SNHL when exposed to OM compared to controls (healthy "volunteers" of same age and gender)[27] in case-control studies
BC = Bone Conduction, CI = Confidence Interval, dB = decibel, kHz = kilo Hertz, OR = Odds Ratio
** Odds ratio was calculated manually using the reported data of the study by authors (H.E.; H.v.O.)*

Table 3.3: Results of studies included in Systematic Review: Mean and median BC thresholds of the OM-group and the contralateral ear
Relevant results of BC thresholds reported in mean and/or median of cohort studies included in Systematic Review are presented.
dB = decibel, Hz = Hertz OM = otitis media, Min = Minimum, Max = Maximum, SD = Standard Deviation

Table 3.4: Results of studies included in Systematic Review: Mean and median BC thresholds of the OM-group and the control-group
Relevant results of BC thresholds reported in mean and/or median of case-control studies included in Systematic Review are presented.
dB = decibel, Hz = Hertz OM = otitis media, Min = Minimum, Max = Maximum, SD = Standard Deviation

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Table 1: Risk of bias assessment of included studies according to the Newcastle-Ottawa Quality Assessment Scale

Author (year) / Domain:	Amali (2017)	De Azevedo (2007)	da Costa (2009)	Kolo (2011)	Krakau (2017)	Paparella (1984)	Papp (2003)	Rana (2019)	Subramaniam (2020)
Selection *Max: 4* ⋆	**3** ⋆	**3** ⋆	**3** ⋆	**3** ⋆	**2** ⋆	**3** ⋆	**2** ⋆	**3** ⋆	**3** ⋆
Comparability *Max: 2* ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆
Outcome *Max: 3* ⋆	**2** ⋆	**2** ⋆	**2** ⋆	**1** ⋆	**2** ⋆	**1** ⋆	**2** ⋆	**3** ⋆	**2** ⋆
Quality assessment	**G**	**G**	**G**	**P**	**F**	**P**	**F**	**G**	**G**

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Table 2: Characteristics of studies included in Systematic Review

Author (country, year of publication)	Study design	Sample size OM group (control group)	Control group	Mean duration of disease in years ±SD (min-max)	Mean age in years ±SD (min-max)	OM-group definition	SNHL outcome definition
Amali (Iran, 2017)	RCS	70 (70)	CE	6.5±7.1 (0.25-30)	37.9±15 (0.25-30)	Unilateral otorrhea >3 months and TMP	Difference in BC threshold for cases and controls
de Azevedo (Brazil, 2007)	RCS	115 (115)	CE	12.4±10.9 (1-42)	26.3±15.7 (4-59)	Chronic unilateral otorrhea	BC loss of ≥30dB
da Costa (Brazil, 2009)	RCS	150*(150)	CE	NR	22±13 (3-55)	Inflammation of MEC >3 months	Difference in BC threshold for cases and controls
Kolo (Nigeria, 2011)	RCS	69 (69)	CE	6.1±6.4 (0.2-20)	28.9±18.6 (4-73)	Unilateral otorrhea >2 months and TMP	Difference in BC threshold for cases and controls
Krakau (Sweden, 2017)	Case-Control	13 (15)	<2 AOM episodes until 30 years of age	***	30.1–31.8 ***	rAOM: ≥3 episodes of AOM during six months or ≥4 episodes during 12 months	NR
*Paparella (Multicenter** *, 1984)*	CS	607 (534)	CE	NR	NR	History of recurrent otorrhea <3 years of selection date	BC loss of ≥15dB, ≥30dB, or any loss at 4kHz
Papp (Hungary, 2003)	RCS	121 (121)	CE ****	10.0±9.6	36.9±14.3	NR	Difference in BC threshold for cases and controls
Rana (India, 2019)	PCS	146 (146)	CE	NR	41.5±8.9 (10-60)	(Unilateral) otorrhea >3 months and TMP	BC threshold of ≥25 dB
Subramaniam (Iran, 2020)	Case-Control	137 (137)	Volunteers of same age and gender	NR	(11-45)	Chronic inflammation of MEC with recurrent episodes of otorrhea through a TMP	BC loss of ≥20 dB at 0.25-4 kHz*****

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Table 3.1: Results of studies included in Systematic Review: Odds ratio of having SNHL when exposed to OM in cohort studies

Author (year)	Control group	SNHL outcome description	OR (95% CI)	P-value
De Azevedo* (2007)	CE	BC loss of ≥30dB	3.30 (1.16-9.40)	0.03
Paparella (1984)	CE	BC loss ≥15dB at ≥2 frequencies BC loss ≥30dB at ≥2 frequencies BC loss a 4kHz only	7.86 (NR) 6.49 (NR) 4.41 (NR)	<0.05 <0.05 <0.05

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Table 3.2: Results of studies included in Systematic Review: Odds ratio of having SNHL when exposed to OM in case-control studies

Author (year)	Control group	SNHL outcome description	OR (95% CI)	P-value
Subramaniam* (2020)	Volunteers of same age and gender	BC loss of ≥20 dB at 0.25-4 kHz	0.05 (0.00-0.78)	0.03

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Table 3.3: Results of studies included in Systematic Review: Mean and median BC thresholds of the OM-group and the contralateral ear

Author (year)	Effect estimate reported in dB: Mean ± SD Median (min/max)		Frequency						
			0.25 kHz	0.5 kHz	1 kHz	1.5 kHz	2 kHz	3 kHz	4 kHz
Amali (2017)	Mean	OM-Group	-	17.57 ± 13.93	18.36 ± 16.30	-	21.71 ± 16.92	-	21.71 ± 16.92
		Control-group	-	10.57 ± 9.07	10.81 ± 9.96	-	13.21 ± 12.10	-	18.29 ± 14.77
		P-value	-	<0.001	<0.001	-	<0.001	-	<0.001
da Costa (2009)	Mean	OM-Group	-	6.63 ± 9.06	6.88 ± 9.01	-	8.81 ± 8.28	12.44 ± 9.41	12.56 ± 10.40
		Control-group	-	5.44 ± 5.69	4.69 ± 4.80	-	5.25 ± 4.96	6.00 ± 4.73	5.69 ± 5.20
		P-value	-	0.001	0.001	-	<0.0001	<0.0001	<0.0001
	Median	OM-Group		5 (0/50)	10 (0/50)	-	10 (0/40)	15 (0/40)	15 (0/50)
		Control-group		5 (0/20)	5 (0/15)	-	5 (0/20)	5 (0/20)	5 (0/20)
Kolo (2011)	Mean	OM-Group	-	32.46 ± 14.587	36.06 ± 14.140	-	42.18 ± 15.183	-	45.511 ± 15.384
		Control-group	-	10.14 ± 4.470	10.14 ±5.208	-	9.97 ± 4.585	-	10.99 ± 3.930
		P-value	-	<0.05	<0.05	-	<0.05	-	<0.05
Papp (2003)	Mean	OM-Group	-	10.4 ± 4.89				-	11.9 ± 6.78
		Control-group	-	4.49 ± 3.947				-	3.10 ± 4.510
		P-value	-	NR				-	NR
	Median	OM-Group		10 (3.3/30)				-	3.34 (0/21.67)
		Control-group		10 (2.5/35)				-	2.5 (0/30)
Rana (2019)	Mean	OM-Group	-	23.50 ± 4.5	24.06 ± 2.4	-	36.18 ± 5.6	-	40.50 ± 6.4
		Control-group	-	18.40 (SD 4.2)	16.6 ± 2.1	-	9.90 ± 4.1	-	11.00 ± 5.2
		P-value	-	<0.001	<0.001	-	<0.001	-	<0.001

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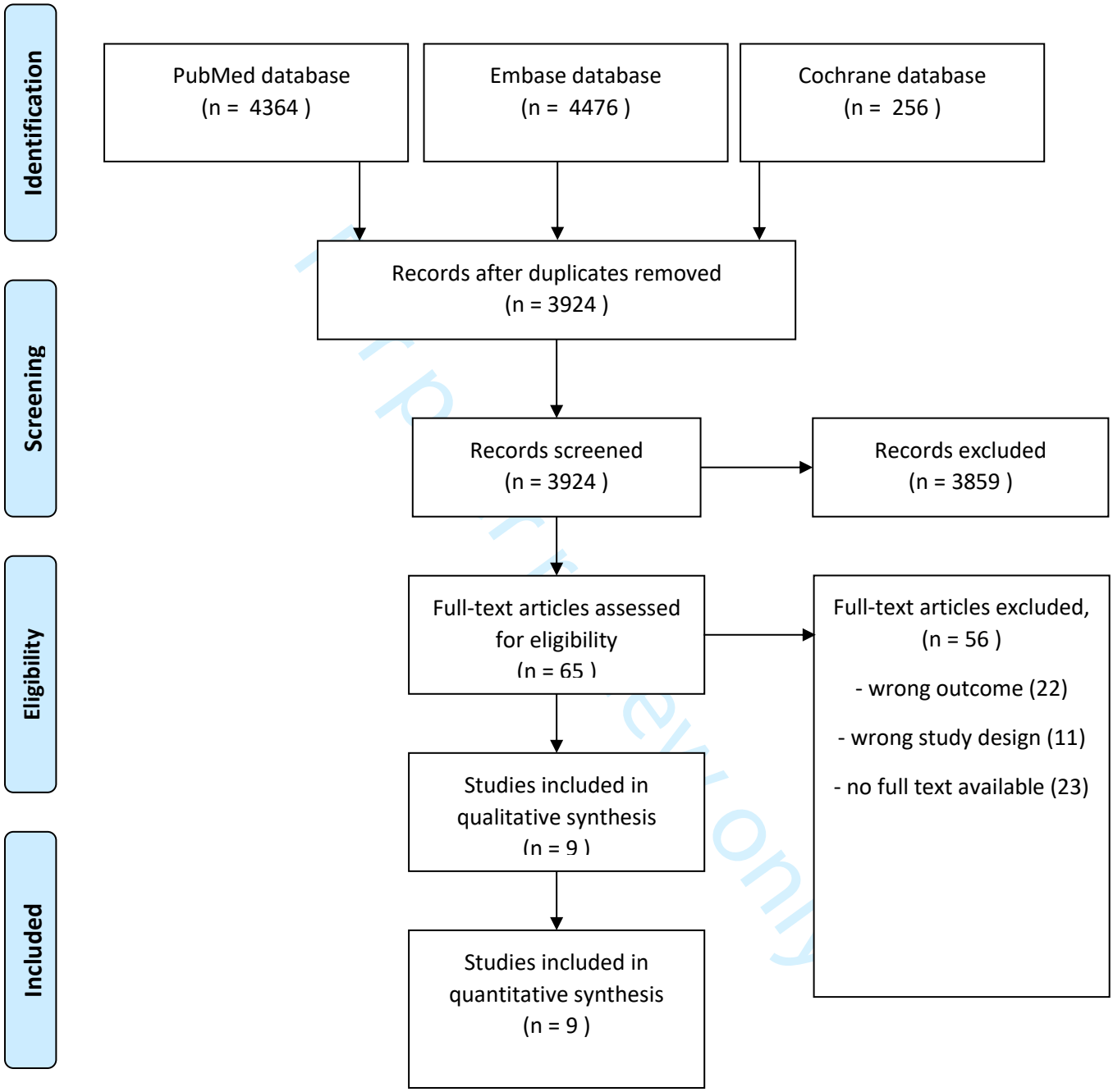
Table 3.4: Results of studies included in Systematic Review: Mean and median BC thresholds of the OM-group and the control-group

Author (year)	Effect estimate reported in dB: Mean ± SD Median (min/max)		Frequency						
			0.25 kHz	0.5 kHz	1 kHz	1.5 kHz	2 kHz	3 kHz	4 kHz
Krakau (2017)	Median	OM-Group Right ear	5 (0 /20)	5 (0/15)	0 (-5/10)	0 (-10/25)	10 (0/20)	10 (-5/25)	0 (-5/25)
		Control-group Right ear	0 (-5/10)	5 (-5/10)	0 (-10/10)	0 (-10/15)	10 (-10/40)	0 (-5/10)	5 (-10/20)
		OM-Group Left ear	5 (0/20)	5 (0/20)	0 (-10/10)	0 (-10/10)	10 (-5/45)	5 (-5/25)	5 (-5/25)
		Control-group Left ear	5 (-5/10)	5 (0/15)	0 (-5/10)	5 (-5/20)	10 (0/45)	0 (-5/10)	0 (-10/15)
		P-value	NR						

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Figure 1: Flow chart for included studies



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APPENDIX 1: Search syntax

PUBMED SEARCH
"Otitis Media"[Mesh] OR otitis media[Title/Abstract] OR ear infection*[Title/Abstract] OR ear inflammation*[Title/Abstract] OR otiti* media[Title/Abstract] OR middle ear inf*[Title/Abstract] OR OME[Title/Abstract] OR OMA[Title/Abstract] OR CSOM[Title/Abstract] OR otitides[Title/Abstract] OR ACSOM[Title/Abstract] AND "Hearing Loss"[Mesh] OR hearing loss*[Title/Abstract] OR deafness[Title/Abstract] OR cochlear dysfunction*[Title/Abstract] OR hypoacusis[Title/Abstract] OR hearing impairment[Title/Abstract] OR inner ear impairment[Title/Abstract]
EMBASE SEARCH
'otitis media'/exp OR 'otitis media':ti,ab,kw OR 'ear infection*':ti,ab,kw OR 'ear inflammation*':ti,ab,kw OR 'otiti* media':ti,ab,kw OR 'middle ear inf*':ti,ab,kw OR 'OME':ti,ab,kw OR 'OMA':ti,ab,kw OR 'CSOM':ti,ab,kw OR 'otitides':ti,ab,kw OR 'ACSOM':ti,ab,kw AND 'inner ear hearing loss'/exp OR 'hearing loss*':ti,ab,kw OR 'deafness':ti,ab,kw OR 'cochlear dysfunction*':ti,ab,kw OR 'hypoacusis':ti,ab,kw OR 'hearing impairment':ti,ab,kw OR 'inner ear impairment':ti,ab,kw
COCHRANE SEARCH
"otitis media":ti,ab,kw OR "ear infection*":ti,ab,kw OR "ear inflammation*":ti,ab,kw OR "otiti* media":ti,ab,kw OR "middle ear inf*":ti,ab,kw OR "OME":ti,ab,kw OR "OMA":ti,ab,kw OR "CSOM":ti,ab,kw OR "otitides":ti,ab,kw OR "ACSOM":ti,ab,kw AND "hearing loss*":ti,ab,kw OR "deafness":ti,ab,kw OR "cochlear dysfunction*":ti,ab,kw OR "hypoacusis":ti,ab,kw OR "hearing impairment":ti,ab,kw OR "inner ear impairment":ti,ab,kw

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APPENDIX 2: predefined risk of bias consistently criteria

The Quality In Prognosis Studies tool (QUIPS)

Domain	Criteria
Study Participation	- In-/exclusion criteria: if only including patients with audiograms, choose no (selection bias). - Baseline characteristics: when most important factors are described but no table is given, choose yes.
Study attrition	No additional criteria
Prognostic factor measurement	- Definition of the prognostic factor: must contain >3 months.
Outcome measurement	No additional criteria
Study confounding	- Confounders: age, duration of disease, (prior) use of ototoxic ear drugs.
Statistical analysis and reporting	No additional criteria

Newcastle-Ottawa Score (NOS)

Category	Criteria
• Selection	No additional criteria
• Comparability	- Total of 1 star when accounting for age (cohort: contralateral ear; case-control: matching). - Total of 2 stars when also accounting for duration of disease and prior use of ear drugs.
• Outcome	- Choose record linkage when outcome information was derived from chart reviews. - Follow up was long enough when duration of disease was >3 months.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	5
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7-8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	8-9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	8-9
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	8-9
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	10

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Relation between otitis media and sensorineural hearing loss: a systematic review

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Title:

Relation between otitis media and sensorineural hearing loss: a systematic review

Authors:
Hester B.E. Elzinga[1,*] & Hanneke D. van Oorschot[1,*] & Inge Stegeman [1,2, 3, 4] & Adriana L. Smit [1,2]

Corresponding author:

A.L. Smit, MD, PhD.

Department of Otorhinolaryngology and Head & Neck Surgery, P.O. Box 85500, 3508 GA Utrecht,

University Medical Center Utrecht, Utrecht, The Netherlands. E: a.l.smit-9@umcutrecht.nl / T: 0031-

88-7573291

Author affiliations:

1 Department of Otorhinolaryngology, Head and Neck Surgery, University Medical Center Utrecht, Utrecht, The Netherlands

2 Brain Center, University Medical Center Utrecht, Utrecht, The Netherlands

3 Department of Ophthalmology, University Medical Center Utrecht, Utrecht, The Netherlands

4 Epidemiology and Data Science, Amsterdam University Medical Centers, University of Amsterdam, Amsterdam, Netherlands.

*Both authors contributed equally to first authorship

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ABSTRACT

Objectives:

This systematic review summarizes the evidence on the correlation between recurrent acute otitis media (rAOM) or chronic suppurative otitis media (CSOM) and sensorineural hearing loss (SNHL).

Research methods:

PubMed, Embase and Cochrane Library databases were searched from inception to 15-January-2021. Two authors independently identified articles, extracted data and performed quality assessment for included studies. Studies comparing the sensorineural hearing levels of patients with a history of rAOM/CSOM for >3 months to a control-group were included.

Results:

Screening of 4168 articles lead to inclusion of two case-control studies (control-group: non-OM patients) and seven cohort-studies (control-group: contralateral ear). Quality assessment indicated considerable risk of bias in all studies. Reported populations varied (sample size 13-607, mean age 22-41.5 years, mean duration of disease 6.1-12.4 years). The odds ratio for SNHL in the OM-group was 3.30-7.86 (95%CI 1.16-9.40, $p<0.05$) in cohort studies (n=2), and 0.05 (95%CI 0-0.78, $p<0.05$) in a case-control study. Mean/median bone conduction thresholds were respectively 1.19-32.21/0-10 dB higher on all frequencies (0.5-4kHz) for the OM-group in four cohort studies ($p<0.05$). Two other studies reported no statistical test outcomes.

Conclusion:

Due to the high risk of bias of included studies, effect estimates heterogeneity, and suboptimal research designs, no conclusion on the correlation between OM and SNHL can be made. It emphasizes the need for future prognostic studies.

Abstract word count: 216 words

Keywords: otitis media, OM, sensorineural hearing loss, SNHL, systematic review

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- A minimal otitis-exposure of three months was adhered to safeguard presence of inner-ear toxin-exposure before outcome assessment.
- A lack of prognostic studies complicates interpretation of effect estimates.
- Heterogeneity of definitions for otitis media and sensorineural hearing loss amongst studies complicate comparability.
- Recruitment of patients in an otolaryngology-setting with audiogram data introduces selection bias.
- Deficits in longitudinal design and methodology limit the quality of our results.

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INTRODUCTION

Otitis media (OM) has a great variety in presentation; it can occur transient as acute otitis media (AOM) and otitis media with effusion, or persistent as recurrent acute otitis media (rAOM) and chronic suppurative otitis media (CSOM).[1] It impacts lives worldwide, with global incidence rates estimated at 11% for AOM and 5% for CSOM, of which respectively 51% and 23% are children.[2] The exact incidences of the OM-subtypes vary between populations, depending on social, demographic and genetic factors for instance.[1]

During the acute inflammatory process, OM can lead to conductive hearing loss (HL).[3] In children this transient loss during the acute phase has not been related to a significant risk of delayed speech development or impaired school performance later on.[4,5] Though, the inflammatory process of the middle ear can be detrimental to the inner ear as shown in histological and animal studies.[6–8] During the inflammatory process, toxins can initiate damage to the round window membrane. This leads to diffusion of toxins to the inner ear and leakage of inner ear fluids, resulting in cochlear function loss; a permanent loss.[6–8] Till date, it remains unknown how many episodes of OM, and what duration of disease lead to inner ear hearing loss.

Given the age-related deterioration of inner-ear function, additional OM-related cochlear damage may become more prominent in later life. In observational studies, a statistical significant higher risk of developing permanent conductive and/or sensorineural hearing loss (SNHL) in adulthood has been demonstrated after experiencing CSOM at childhood age.[9–11] Several studies reported incidence numbers of SNHL up to 23-24% in CSOM patients.[12,13] In contrast to other risk factors for SNHL, OM is considered treatable by antibiotic or surgical interventions, which are available treatment options in most countries.[7] Though, up to this day, consensus on the long term hearing effects of OM has not been met, which complicates drawing conclusions and making recommendation for clinical care. [1,3]

Considering the above, we recognize a need for evidence on the effect of OM on sensorineural hearing levels in time. This knowledge could contribute to establish the best treatment guideline for this common condition, avoiding possible adverse long-term effects. We therefore conducted a systematic review to summarize the evidence on the correlation between rAOM or CSOM and SNHL in time.

METHODS

Methods - Patient and Public Involvement

No patients were involved.

Methods - reporting

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We applied the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline for this systematic review.[14]

Methods - Search strategy

On 15th of January 2021 a systematic search was performed in Embase, PubMed and Cochrane Library databases. No filters were applied. We used search terms and synonyms for the domain (CSOM, rAOM) and outcome (SNHL), including Emtree, and medical subject headings (MeSH) term fields. Used synonyms for the domain were otitis media, (middle) ear infection/inflammation, and abbreviations. Used synonyms for the outcome were hearing loss/impairment, deafness, cochlear dysfunction, hypoacusis, and inner ear impairment (Search syntax in Appendix 1). Although SOM did not apply to the study domain, it was included as a synonym to prevent exclusion of relevant articles during the initial search.

Methods - Study selection

Two authors blindly screened studies for inclusion (H.E., H.v.O.), and disagreements were resolved by discussion. Where disagreements could not be resolved, a third reviewer (D.S.) made the final decision. Prognostic, cohort, and cross-sectional studies were included. Cohort studies were categorized in retrospective or prospective study designs. Where this was methodologically unclear, the study was labeled as cohort study. Conference abstracts, or case reports were excluded. In case of an unavailable full text or abstract, the author was contacted to request full text. Reference screening and backwards citation tracking were performed to identify any missed literature. Rayyan was used as screening tool.[15] Full text screening and data extraction was independently performed by two authors (H.E., H.v.O.).

Original studies reporting on hearing outcome of patients of all ages with rAOM or CSOM and a control-group were included. Studies including participants with SNHL-related syndromes, cholesteatoma, or a history of ear surgery were excluded, as these factors independently cause SNHL. Articles that reported outcomes for several subgroups, of which only a selection met the inclusion criteria, data of that section was extracted and included in the analysis. CSOM was defined as a minimal presence of OM for three consecutive months or longer. Studies that included patients with OM of more than three months or had a minimal follow-up of three months were included. These criteria were set to safeguard a minimum period of three months in which inner-ear toxin exposure was present, before outcomes were assessed. Studies reporting effect estimates on hearing outcome for both the exposed and control-group were selected. To establish the relation between OM and SNHL, a shift of $> \underline{5}$ dB in hearing level between tone audiometric test results was found relevant considering the search range of 5 dB during these test procedures.[16]

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Methods – Data collection and analysis

Quality assessment of the studies

Critical appraisal of included studies was independently performed by two reviewers (H.E., H.v.O.) for validity and relevance. The Quality In Prognosis Studies (QUIPS) tool was used to assess risk of bias for prognostic study designs, and the Newcastle-Ottawa Scale (NOS) was used for cohort and case-control studies.[17,18] To score risk of bias consistently, criteria were predefined for both tools by the authors (Appendix 2). (H.E., H.v.O.) Using these tools, studies could be scored to contain low, moderate or high risk of bias (QUIPS), or to be of good, fair or poor quality (NOS). Differences in the appraisal of studies between the two reviewers in either tool were resolved through discussion.

Data and outcome extraction

Data on study and patient characteristics, follow-up, and BC hearing outcome of the included patients was independently extracted by two reviewers (H.E., H.v.O.). The primary outcome measure of cases and controls was reported in BC thresholds (dB) per frequency (0.5-4 kHz) and/or average hearing levels. Age, duration of disease, and the use of ototoxic ear drops were considered as confounding factors for outcomes of SNHL in groups.

Outcome measures and analyses

Odds ratios (OR) were either extracted directly or calculated by the authors (H.E., H.v.O.) if data was sufficient. If an OR could not be extracted or calculated, means and medians were extracted. A p-value of <0.05 was considered statistically significant. A meta-analysis was performed in case of methodologically homogenous studies.

RESULTS

Results - Search strategy and study collection

The search identified 4168 articles (Figure 1). After removal of duplicates, 3725 (89.4%) studies were screened based on title and abstract, as a result of which 3662 (98.3%) studies were excluded. After full-text analysis, 54 articles were excluded based on the outcome (n=22), study design (n=9), or by unavailability of the full text (n=23). No articles were received after correspondence with authors for full text. This resulted in a total of nine articles meeting all inclusion criteria.[19–27] Seven studies were cohort studies with a retrospective or prospective study design, or not further specified

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design.[19,20,22,24–26,28] Two studies were case-control studies.[23,27] No prognostic studies were found. Due to the methodological heterogeneity in included studies, no meta-analysis could be performed.

[INSERT FIGURE 1]

Results - Quality of the included articles

Critical appraisal of all studies was performed with the NOS (Table 1). The quality of included studies ranged from good (n=5),[19,20,26–28] to fair (n=2),[23,25] and poor (n=2).[22,24]

Selection

Selection bias was present in all studies, as patients were visiting an otolaryngology department before inclusion, none of the studies included a longitudinal cohort of the general population. Also, there was no objective demonstration that SNHL was not present before inclusion.[19–27] One study included self-reported diagnosis of OM instead of a doctor diagnosis.[23] Another study did not document information on recruitment of the cohort.[25]

Comparability

Most studies excluded patients with a history of ototoxic drug use.[19,22,24–27] Case-control studies matched cases and controls according to age and gender,[23,27] cohort studies used the contralateral ear as control-group.[19–22,24–26] Hearing outcomes were not corrected for known or unknow confounding factors in the included studies.[19–27]

Outcome

Rana et al. was the only study scoring low risk of bias for the outcome domain.[26] All other studies did not report information on missing data or loss to follow up.[19,20,23,25,27,28] Two studies could not guarantee a minimum duration of disease of three months for all their included patients.[22,24] Demographic information of included patients was retrieved from chart reviews. Otoscopy and audiometry were performed by trained personnel using conventional equipment.[19–27] Only the study of *de Azevedo* et al. reported that audiometry exams were carried out blinded in double by two independent professionals.[20]

[INSERT TABLE 1]

Results - Data extraction and study characteristics

Trial design and study sample

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Included studies were published between 1984 and 2019 and pertained to patients from Africa, Asia, Europe, North America and South America (Table 2). The sample sizes for the OM-group in cohort and case-control studies ranged from 70-607 (controls 70-534) and 13-137 (controls 13-137) respectively. Cohort studies used the contralateral ear as control-group.[19–22,24–26] Case-control studies used adults with <2 AOM episodes until 30 years of age[23] and volunteers of same age and gender[27] as control-group. Mean duration of disease varied between 6.1-12.4 years (range 0.2-42 years) in the four studies reporting it.[19,20,22,25] The mean age of patients was 22-41.5 years (range 0.25-73 years). Most study populations included both adults and children, except for two studies.[23,25] *Paparella* et al. did not describe the age of included patients, but excluded patients older than 60 years.[24]

Definitions for OM and SNHL

The included studies handled different inclusion criteria for the OM-group, namely: unilateral otorrhea >2 or >3 months with a tympanic membrane perforation (TMP)[19,22,26], chronic unilateral otorrhea[20], inflammation of the middle ear compartment (MEC) of >3 months[21], rAOM defined as ≥3 episodes of AOM during six months or ≥4 episodes during 12 months[23], a history of recurrent otorrhea within 3 years before data assembly[24], and chronic inflammation of the MEC with recurrent episodes of otorrhea through a TMP[27]. For *Krakau* et al., no criteria for inclusion in the OM-group were reported.[25]

Four studies defined their outcome as difference in BC threshold (dB) for cases and controls.[19,21,22,25] Four other studies defined a specific BC threshold level to establish cases with SNHL, namely: ≥30dB,[20] ≥15dB/≥30dB/any loss at 4 kHz,[24] ≥25dB,[26] and ≥20dB.[27] *Papp* et al. did not report a hearing outcome definition to be considered as SNHL.[23]

[INSERT TABLE 2]

Results - Outcomes

Three studies (2 cohort studies, 1 case-control study) reported odds ratios (OR) for the presence of SNHL after three months or longer exposure to OM.[20,24,27] (Table 3.1 and 3.2) The other six studies (5 cohort studies, 1 case-control studies) reported their outcome as a mean and/or median BC threshold for the OM-group and the control-group (Table 3.3 and 3.4).[19,21–23,25,26]

Outcomes in cohort studies

The OR for having SNHL could be extracted or calculated in two cohort studies comparing the contralateral to the diseased ear (Table 3.1).[20,24] Both reported higher ORs for the presence of SNHL after three months or longer exposure to OM. *Paparella* et al. reported an OR of 7.86 for a BC loss of

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≥15dB at ≥2 frequencies, 6.49 for BC loss of ≥30dB at ≥2 frequencies, and 4.41 for any BC loss at 4kHz (95% CI not reported).[24] For *de Azevedo* et al., the OR was 3.30, (95% CI 1.16-9.40, p=0.03).[20]

Five cohort studies reported a mean,[19,22,26] or both mean and median[21,25] for the BC hearing levels per frequency, stratified per study group (Table 3.3). Of those, four studies had a higher mean and/or median BC threshold in the OM-group compared to the control-group for all individually tested frequencies (0.5-4 kHz).[19,21,22,26] Mean and/or median differences between both groups per frequency ranged from 1.19-32.21 dB (p-value <0.05) for means and 0-10 dB (p-value <0.05) for medians.[19,21,22,26] *Papp* et al. reported the mean and median BC threshold for low (0.5-2kHz) and high (4 kHz) frequencies. The mean differences were higher for the OM-group compared to the control-group, though no statistical significance was determined.[25]

Outcomes in case-control studies

Subramaniam et al. reported data that was sufficient to calculate the OR (Table 3.2: 0.05 (95% CI 0.00-0.78, p=0.03) for SNHL when comparing the OM-patients to healthy volunteers, with SNHL defined as a BC loss of ≥20 dB at 0.25-4 Hz (Table 3.2).[27]

In one case-control study by *Krakau* et al. the authors reported median BC thresholds for the OM and control-group, stratified for the left and right ear (Table 3.4). The right ear showed a higher BC threshold (difference 10 dB, no p-value) for the OM-group compared to the control-group at 3 kHz. For all other frequencies, of both the right and left ear, the BC threshold difference between the groups was 5 dB or less.[23]

[INSERT TABLE 3.1, 3.2, 3.3, 3.4]

DISCUSSION

In this study, we described the results of a systematic search on the relation between rAOM or CSOM and the occurrence of SNHL in time. Two case-control studies (control-group: non-OM patients) and seven cohort-studies (control-group: contralateral ear) reported hearing outcomes for both the exposed and control-group.[19–26] The OR for the presence of SNHL after a minimal exposure to OM of three months was statistically significantly higher in two cohort studies,[20,24] but lower in one case-control study.[27] Studies in which no OR was provided or could be calculated, mean and/or median BC thresholds were statistically significantly higher in the OM-group in four out of six studies. Mean or median differences between both groups ranged between 1.19-32.21 dB and 0-10 dB respectively.[19,21,22,26] The other two studies reported differences of 0-5 dB in mean or median between OM-group and the contralateral ear[25] or controls, [23] though did not provide outcomes of statistical

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testing between groups. Outcomes of included studies should be interpreted cautiously by the small shift in hearing thresholds observed. Besides this, the heterogeneity between studies, the limited quality of included studies, and the lack of prognostic studies are all major limitations to draw conclusions about the effect of OM on SNHL.

None of the included articles assessed the effect of duration of the OM on SNHL. A longer period of exposure of the inner ear to inflammation, and therefore to toxins, could increase the portion of patients with SNHL.[29] Previous population studies analyzed the effect of the duration of the OM on hearing levels. *Thakur* et al. studied 100 cases (15-50 years of age) presenting with unilateral mucosal COM. They found an SNHL incidence of 23% (hearing level >20 dB), with a statistically significant positive correlation between duration of disease and SNHL.[12] A comparable SNHL incidence of 24% in 100 unilateral CSOM cases (aged 11-50 years) was published by *Kaur* et al, where the SNHL incidence (defined as hearing level >20 dB) increased with the duration of disease (up to >30 years).[13]

Besides exposure-time, one should take the following into consideration. Included studies were heterogenous due to varying sample sizes, age of study populations, definitions used for the OM-group, criteria for SNHL and method of reporting of outcome. Recruitment of patients was mostly done in an ENT-setting, and some studies handled absence of an audiogram as exclusion criterion which introduces selection or information bias. For example, *Kolo* et al. excluded patients if they had "incomplete" clinical records or no audiograms.[22] Age was accounted for by matching cases and controls or by using the contralateral ear. The use of the contralateral ear as non-affected control-group in cohort studies had the advantage of limiting the effect of inter-subject variations. On the other hand, if this ear suffered from infection prior to the study start, but was free of disease during the inclusion process, this could influence outcomes. However, in case-control studies, it could not be guaranteed that their control-group was free of previous OM-exposure.[22,24] For instance, *Krakau* et al. included a patient with 10 self-reported episodes of AOM in the control-group.[23] As stated above, no prognostic studies were included, and a meta-analysis could not be performed due to heterogeneity of reported results. A limitation of the present study is the considerable number of studies of which the full text could not be retrieved.

One could hypothesize that when inflammatory toxins are detrimental to the inner ear, the higher frequencies could be most affected considering their location at the base of the cochlea and in proximity of the inflammatory process in the middle ear. Four out of seven of our included studies reporting mean and/or median BC thresholds show a larger difference in hearing loss in the higher frequencies (up to 4 kHz). This is of interest for further studies as it could make people prone for hearing loss in the middle frequency by progression of a high frequency hearing loss by ageing.

The lack of longitudinal study designs and the selected sample of population studied so far hinders strong conclusions. In order to overcome this methodological flaw, studies with a longitudinal

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design, assessing a random sample of children and adults with and without OM for several decades is necessary.[30]

CONCLUSION

Considering the high risk of bias and the suboptimal research design of current evidence, no conclusions could be made about the relation between otitis media and SNHL in time. The high impact of SNHL on daily life, the high prevalence of OM in children and adults, and its potential effect on hearing, underpin the need for longitudinal studies assessing the effect of OM on hearing loss in time.

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AUTHOR CONTRIBUTIONS

I.S. and A.S. were involved in conception of the study. H.E. and H.v.O. drafted the manuscript, searched and reviewed literature, assessed the quality of the studies, extracted data, and performed statistical analysis. H.E., H.v.O., I.S. and A.S. contributed to data review and interpretation and manuscript review. H.E., H.v.O., I.S. and A.S. have read and approved the final manuscript.

DATA STATEMENT

All data relevant to the study are included in the article or uploaded as supplementary information.

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ETHICS STATEMENT

Authors declare there was no conflict of interest. This article does not contain any studies involving human participants or animals performed by any of the authors.

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TABLE AND FIGURE LEGENDS

Figure 1: Flow chart for included studies
Last date of search January 15th 2021 PRISMA 2009 flow diagram.[14]

Table 1: Risk of bias assessment of included studies according to the Newcastle-Ottawa Quality Assessment Scale
Risk of bias assessment of included studies according to the Newcastle-Ottawa Quality Assessment Scale[17]
F = Fair quality, G = Good quality, Max = maximum, P = Poor quality

Table 2: Characteristics of studies included in Systematic Review
The characteristics of the included studies are presented.
AOM = Acute Otitis Media, BC = Bone Conduction, BCT = Bone Conduction Threshold, CE = contralateral ear, COM = Chronic Otitis Media, CS = Cohort Study, dB = decibel, Hz = hertz, MEC = Middle Ear Compartment, OM = Otitis Media, PCS = Prospective Cohort Study, rAOM = recurrent Acute Otitis Media, RCS = Retrospective Cohort Study, SD = standard deviation, SNHL = Sensorineural Hearing Loss, TMP = Tympanic Membrane Perforation. NR = Not Reported
** Including 31% patients with cholesteatoma.*
*** Multicenter study reporting on subjects from Minnesota, Italy, Finland, Sweden, Korea and New York*[24]
****Research population +- 30 years of age with a history of rAOM during childhood*[23]
*****Averages in BC threshold of the contralateral (normal) ear were subtracted from that of the CSOM side.*[25]
****** The obtained thresholds were used to quantitatively assess the degree of hearing loss using Clark's (1981) modification of Goodman classification of severity of hearing loss (1965)*[27]

Table 3.1: Results of studies included in Systematic Review: Odds ratio of having SNHL when exposed to OM in cohort studies
Relevant results reported in odds ratio for having SNHL when exposed to OM compared to controls (CE)[20,24] in cohort studies
BC = Bone Conduction, CE = Contralateral Ear, CI = Confidence Interval, dB = decibel, kHz = kilo Hertz, OR = Odds Ratio
** Odds ratio was calculated manually using the reported data of the study by authors (H.E.; H.v.O.)*

Table 3.2: Results of studies included in Systematic Review: Odds ratio of having SNHL when exposed to OM in case-control studies
Relevant results reported in odds ratio for having SNHL when exposed to OM compared to controls (healthy "volunteers" of same age and gender)[27] in case-control studies
BC = Bone Conduction, CI = Confidence Interval, dB = decibel, kHz = kilo Hertz, OR = Odds Ratio
** Odds ratio was calculated manually using the reported data of the study by authors (H.E.; H.v.O.)*

Table 3.3: Results of studies included in Systematic Review: Mean and median BC thresholds of the OM-group and the contralateral ear
Relevant results of BC thresholds reported in mean and/or median of cohort studies included in Systematic Review are presented.
dB = decibel, Hz = Hertz OM = otitis media, Min = Minimum, Max = Maximum, SD = Standard Deviation

Table 3.4: Results of studies included in Systematic Review: Mean and median BC thresholds of the OM-group and the control-group
Relevant results of BC thresholds reported in mean and/or median of case-control studies included in Systematic Review are presented.
dB = decibel, Hz = Hertz OM = otitis media, Min = Minimum, Max = Maximum, SD = Standard Deviation

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Table 1: Risk of bias assessment of included studies according to the Newcastle-Ottawa Quality Assessment Scale

Author (year) / Domain:	Amali (2017)	De Azevedo (2007)	da Costa (2009)	Kolo (2011)	Krakau (2017)	Paparella (1984)	Papp (2003)	Rana (2019)	Subramaniam (2020)
Selection *Max: 4* ⋆	**3** ⋆	**3** ⋆	**3** ⋆	**3** ⋆	**2** ⋆	**3** ⋆	**2** ⋆	**3** ⋆	**3** ⋆
Comparability *Max: 2* ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆	**1** ⋆
Outcome *Max: 3* ⋆	**2** ⋆	**2** ⋆	**2** ⋆	**1** ⋆	**2** ⋆	**1** ⋆	**2** ⋆	**3** ⋆	**2** ⋆
Quality assessment	**G**	**G**	**G**	**P**	**F**	**P**	**F**	**G**	**G**

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Table 2: Characteristics of studies included in Systematic Review

Author (country, year of publication)	Study design	Sample size OM group (control group)	Control group	Mean duration of disease in years ±SD (min-max)	Mean age in years ±SD (min-max)	OM-group definition	SNHL outcome definition
Amali (Iran, 2017)	RCS	70 (70)	CE	6.5±7.1 (0.25-30)	37.9±15 (0.25-30)	Unilateral otorrhea >3 months and TMP	Difference in BC threshold for cases and controls
de Azevedo (Brazil, 2007)	RCS	115 (115)	CE	12.4±10.9 (1-42)	26.3±15.7 (4-59)	Chronic unilateral otorrhea	BC loss of ≥30dB
da Costa (Brazil, 2009)	RCS	150*(150)	CE	NR	22±13 (3-55)	Inflammation of MEC >3 months	Difference in BC threshold for cases and controls
Kolo (Nigeria, 2011)	RCS	69 (69)	CE	6.1±6.4 (0.2-20)	28.9±18.6 (4-73)	Unilateral otorrhea >2 months and TMP	Difference in BC threshold for cases and controls
Krakau (Sweden, 2017)	Case-Control	13 (15)	<2 AOM episodes until 30 years of age	***	30.1–31.8 ***	rAOM: ≥3 episodes of AOM during six months or ≥4 episodes during 12 months	NR
*Paparella (Multicenter** *, 1984)*	CS	607 (534)	CE	NR	NR	History of recurrent otorrhea <3 years of selection date	BC loss of ≥15dB, ≥30dB, or any loss at 4kHz
Papp (Hungary, 2003)	RCS	121 (121)	CE ****	10.0±9.6	36.9±14.3	NR	Difference in BC threshold for cases and controls
Rana (India, 2019)	PCS	146 (146)	CE	NR	41.5±8.9 (10-60)	(Unilateral) otorrhea >3 months and TMP	BC threshold of ≥25 dB
Subramaniam (Iran, 2020)	Case-Control	137 (137)	Volunteers of same age and gender	NR	(11-45)	Chronic inflammation of MEC with recurrent episodes of otorrhea through a TMP	BC loss of ≥20 dB at 0.25-4 kHz*****

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Table 3.1: Results of studies included in Systematic Review: Odds ratio of having SNHL when exposed to OM in cohort studies

Author (year)	Control group	SNHL outcome description	OR (95% CI)	P-value
De Azevedo* (2007)	CE	BC loss of ≥30dB	3.30 (1.16-9.40)	0.03
Paparella (1984)	CE	BC loss ≥15dB at ≥2 frequencies BC loss ≥30dB at ≥2 frequencies BC loss a 4kHz only	7.86 (NR) 6.49 (NR) 4.41 (NR)	<0.05 <0.05 <0.05

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Table 3.2: Results of studies included in Systematic Review: Odds ratio of having SNHL when exposed to OM in case-control studies

Author (year)	Control group	SNHL outcome description	OR (95% CI)	P-value
Subramaniam* (2020)	Volunteers of same age and gender	BC loss of ≥20 dB at 0.25-4 kHz	0.05 (0.00-0.78)	0.03

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Table 3.3: Results of studies included in Systematic Review: Mean and median BC thresholds of the OM-group and the contralateral ear

Author (year)	Effect estimate reported in dB: Mean ± SD Median (min/max)		Frequency						
			0.25 kHz	0.5 kHz	1 kHz	1.5 kHz	2 kHz	3 kHz	4 kHz
Amali (2017)	Mean	OM-Group	-	17.57 ± 13.93	18.36 ± 16.30	-	21.71 ± 16.92	-	21.71 ± 16.92
		Control-group	-	10.57 ± 9.07	10.81 ± 9.96	-	13.21 ± 12.10	-	18.29 ± 14.77
		P-value	-	<0.001	<0.001	-	<0.001	-	<0.001
da Costa (2009)	Mean	OM-Group	-	6.63 ± 9.06	6.88 ± 9.01	-	8.81 ± 8.28	12.44 ± 9.41	12.56 ± 10.40
		Control-group	-	5.44 ± 5.69	4.69 ± 4.80	-	5.25 ± 4.96	6.00 ± 4.73	5.69 ± 5.20
		P-value	-	0.001	0.001	-	<0.0001	<0.0001	<0.0001
	Median	OM-Group		5 (0/50)	10 (0/50)	-	10 (0/40)	15 (0/40)	15 (0/50)
		Control-group		5 (0/20)	5 (0/15)	-	5 (0/20)	5 (0/20)	5 (0/20)
Kolo (2011)	Mean	OM-Group	-	32.46 ± 14.587	36.06 ± 14.140	-	42.18 ± 15.183	-	45.511 ± 15.384
		Control-group	-	10.14 ± 4.470	10.14 ±5.208	-	9.97 ± 4.585	-	10.99 ± 3.930
		P-value	-	<0.05	<0.05	-	<0.05	-	<0.05
Papp (2003)	Mean	OM-Group	-	10.4 ± 4.89				-	11.9 ± 6.78
		Control-group	-	4.49 ± 3.947				-	3.10 ± 4.510
		P-value	-	NR				-	NR
	Median	OM-Group		10 (3.3/30)				-	3.34 (0/21.67)
		Control-group		10 (2.5/35)				-	2.5 (0/30)
Rana (2019)	Mean	OM-Group	-	23.50 ± 4.5	24.06 ± 2.4	-	36.18 ± 5.6	-	40.50 ± 6.4
		Control-group	-	18.40 (SD 4.2)	16.6 ± 2.1	-	9.90 ± 4.1	-	11.00 ± 5.2
		P-value	-	<0.001	<0.001	-	<0.001	-	<0.001

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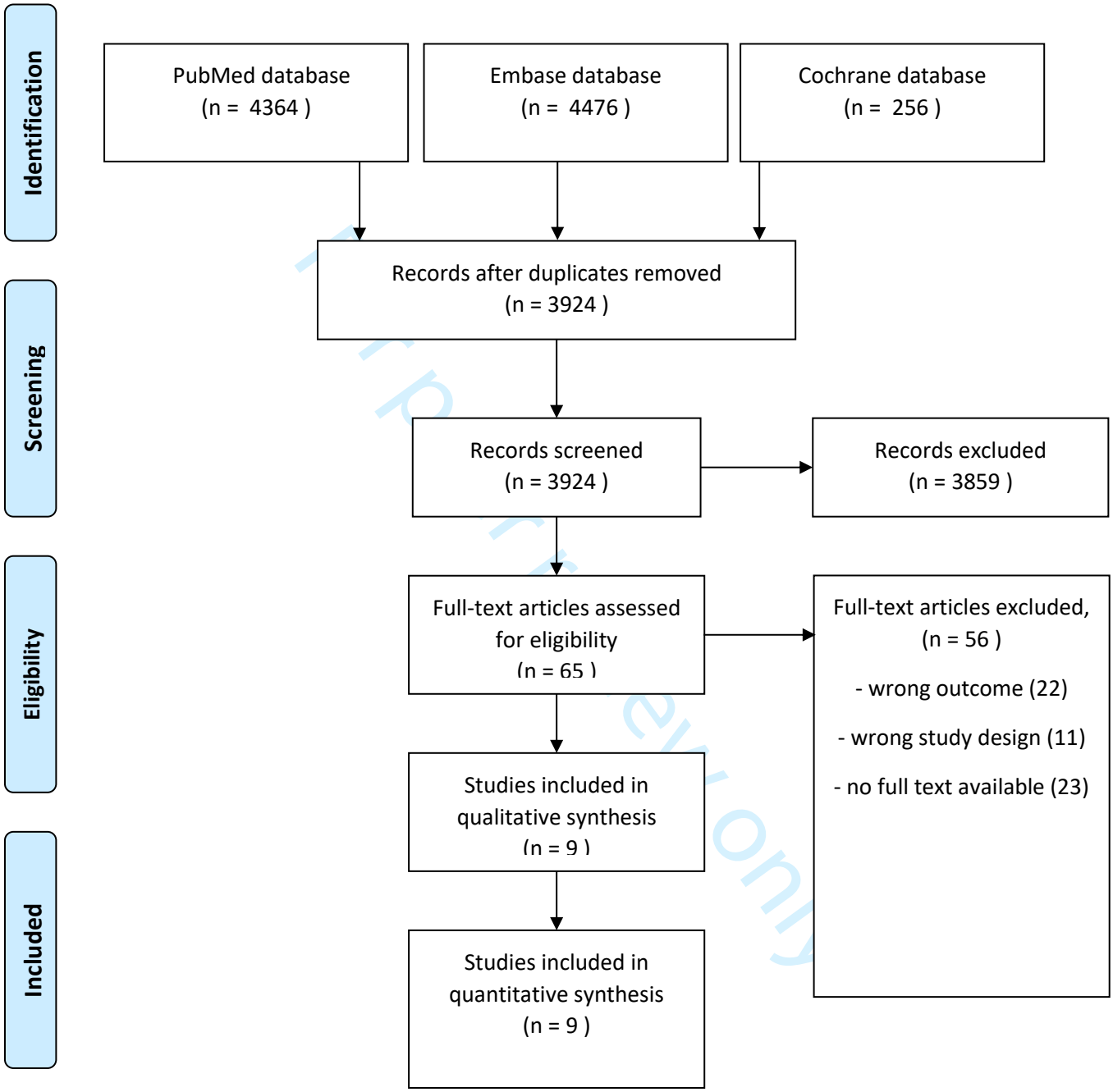
Table 3.4: Results of studies included in Systematic Review: Mean and median BC thresholds of the OM-group and the control-group

Author (year)	Effect estimate reported in dB: Mean ± SD Median (min/max)		Frequency						
			0.25 kHz	0.5 kHz	1 kHz	1.5 kHz	2 kHz	3 kHz	4 kHz
Krakau (2017)	Median	OM-Group Right ear	5 (0 /20)	5 (0/15)	0 (-5/10)	0 (-10/25)	10 (0/20)	10 (-5/25)	0 (-5/25)
		Control-group Right ear	0 (-5/10)	5 (-5/10)	0 (-10/10)	0 (-10/15)	10 (-10/40)	0 (-5/10)	5 (-10/20)
		OM-Group Left ear	5 (0/20)	5 (0/20)	0 (-10/10)	0 (-10/10)	10 (-5/45)	5 (-5/25)	5 (-5/25)
		Control-group Left ear	5 (-5/10)	5 (0/15)	0 (-5/10)	5 (-5/20)	10 (0/45)	0 (-5/10)	0 (-10/15)
		P-value	NR						

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Figure 1: Flow chart for included studies



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APPENDIX 1: Search syntax

PUBMED SEARCH
"Otitis Media"[Mesh] OR otitis media[Title/Abstract] OR ear infection*[Title/Abstract] OR ear inflammation*[Title/Abstract] OR otiti* media[Title/Abstract] OR middle ear inf*[Title/Abstract] OR OME[Title/Abstract] OR OMA[Title/Abstract] OR CSOM[Title/Abstract] OR otitides[Title/Abstract] OR ACSOM[Title/Abstract] AND "Hearing Loss"[Mesh] OR hearing loss*[Title/Abstract] OR deafness[Title/Abstract] OR cochlear dysfunction*[Title/Abstract] OR hypoacusis[Title/Abstract] OR hearing impairment[Title/Abstract] OR inner ear impairment[Title/Abstract]
EMBASE SEARCH
'otitis media'/exp OR 'otitis media':ti,ab,kw OR 'ear infection*':ti,ab,kw OR 'ear inflammation*':ti,ab,kw OR 'otiti* media':ti,ab,kw OR 'middle ear inf*':ti,ab,kw OR 'OME':ti,ab,kw OR 'OMA':ti,ab,kw OR 'CSOM':ti,ab,kw OR 'otitides':ti,ab,kw OR 'ACSOM':ti,ab,kw AND 'inner ear hearing loss'/exp OR 'hearing loss*':ti,ab,kw OR 'deafness':ti,ab,kw OR 'cochlear dysfunction*':ti,ab,kw OR 'hypoacusis':ti,ab,kw OR 'hearing impairment':ti,ab,kw OR 'inner ear impairment':ti,ab,kw
COCHRANE SEARCH
"otitis media":ti,ab,kw OR "ear infection*":ti,ab,kw OR "ear inflammation*":ti,ab,kw OR "otiti* media":ti,ab,kw OR "middle ear inf*":ti,ab,kw OR "OME":ti,ab,kw OR "OMA":ti,ab,kw OR "CSOM":ti,ab,kw OR "otitides":ti,ab,kw OR "ACSOM":ti,ab,kw AND "hearing loss*":ti,ab,kw OR "deafness":ti,ab,kw OR "cochlear dysfunction*":ti,ab,kw OR "hypoacusis":ti,ab,kw OR "hearing impairment":ti,ab,kw OR "inner ear impairment":ti,ab,kw

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APPENDIX 2: predefined risk of bias consistently criteria

The Quality In Prognosis Studies tool (QUIPS)

Domain	Criteria
Study Participation	- In-/exclusion criteria: if only including patients with audiograms, choose no (selection bias). - Baseline characteristics: when most important factors are described but no table is given, choose yes.
Study attrition	No additional criteria
Prognostic factor measurement	- Definition of the prognostic factor: must contain >3 months.
Outcome measurement	No additional criteria
Study confounding	- Confounders: age, duration of disease, (prior) use of ototoxic ear drugs.
Statistical analysis and reporting	No additional criteria

Newcastle-Ottawa Score (NOS)

Category	Criteria
• Selection	No additional criteria
• Comparability	- Total of 1 star when accounting for age (cohort: contralateral ear; case-control: matching). - Total of 2 stars when also accounting for duration of disease and prior use of ear drugs.
• Outcome	- Choose record linkage when outcome information was derived from chart reviews. - Follow up was long enough when duration of disease was >3 months.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	5
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7-8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	8-9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	8-9
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	8-9
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	10

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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